UNIVERSITY BIOINFORMATICS GROUP / STUDY EDITION

Antibiotic Resistance in Escherichia coli

An illustrated comparative genomics study handbook

Investigating the Evolution and Distribution of Antibiotic Resistance Among Selected Escherichia coli Strains Using Comparative Genomics

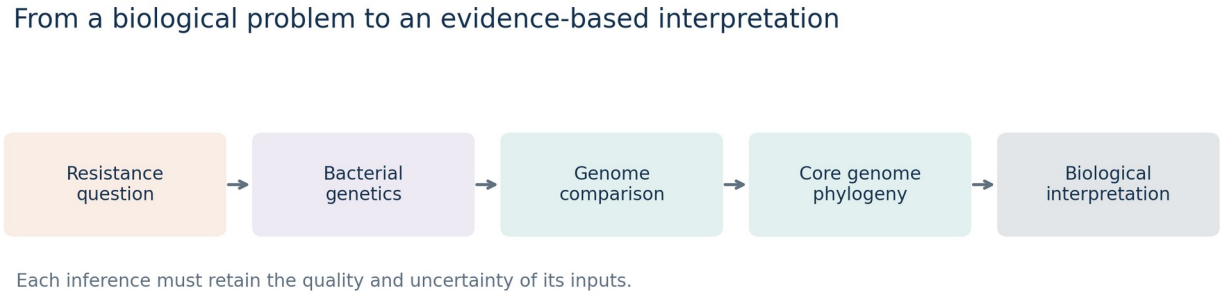


Figure 1 The project logic

Legend and interpretation. Orange marks the problem; purple marks genetic information; teal marks analysis. Follow the arrows from question through genomic evidence to interpretation.

Interpretation limit. This is a proposed reasoning framework, not an analysis result.

Prepared for final-year Informatics and Computer Science students
Track 4 Microbial Bioinformatics / Metagenomics
22 October 2026

Purpose. Build the biological understanding and computational judgement needed to plan, implement and explain a public-genome comparison. This edition contains teaching illustrations and a proposed analysis; it does not report completed genome research.

How to use this handbook

Read Chapters 1–9 to build the biological foundation, Chapters 10–14 to plan the computation, and Chapters 15–17 to practise interpretation. Use the glossary and study questions for group revision. The project can begin with a small pilot before scaling to the agreed collection.

Evidence labels. Established knowledge is explained in prose with relevant citations. Proposed methods describe work the group could perform. Every synthetic sequence, chart, profile and teaching tree is labelled as hypothetical. No public genome accessions, measured resistance phenotypes or experimental results are invented.

Companion materials. The diagram collection contains high-resolution PNG and editable SVG files, the synthetic matrix in CSV, and machine-readable captions. Five GIFs demonstrate selection, plasmid transfer, a nucleotide substitution, branching ancestry and the analysis workflow. A separate three-page summary supports quick group revision.

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In Microsoft Word, right-click the contents and choose Update Field, then Update entire table after making edits. Chapter headings also appear in the Navigation Pane.

Chapter 1 Introduction to the project

The investigation in plain language

Antibiotic resistance occurs when bacteria can withstand an antibiotic that would otherwise inhibit their growth or kill them under defined conditions. The resistant organism is the bacterium, not the person receiving treatment. Our project asks how genetic differences among selected Escherichia coli genomes help explain differences in their predicted resistance potential. We will compare public genome assemblies, identify known resistance determinants, and examine those profiles alongside a tree constructed from shared genomic sequences.

The chosen title is Investigating the Evolution and Distribution of Antibiotic Resistance Among Selected Escherichia coli Strains Using Comparative Genomics. It belongs to Track 4 Microbial Bioinformatics / Metagenomics, but the proposed work is specifically isolate comparative genomics. Metagenomics analyses mixed communities directly; comparing separately assembled isolate genomes does not itself constitute metagenomics.

Research question and objectives

Our central question is: How do antibiotic resistance genes and associated genetic variations differ among selected E. coli strains, and what do these differences reveal about the evolution of antibiotic resistance?

The general objective is to investigate the evolution and distribution of antibiotic resistance among selected E. coli strains using comparative genomic and phylogenetic analysis. The specific objectives are to identify resistance genes; compare their distribution and sequence variation; reconstruct evolutionary relationships using genomic sequence data; and investigate how evolutionary relatedness corresponds to genomic resistance profiles. Unless matched susceptibility measurements are available, the fourth objective concerns predicted genomic profiles rather than measured antibiotic resistance.

Why the question matters

Resistance can make bacterial infections harder to treat and can compromise procedures that depend on effective infection prevention and treatment. Genomic comparison helps distinguish the spread of a successful bacterial lineage from the movement of particular resistance determinants between lineages. These processes can occur together. Understanding them informs surveillance questions, although this small student study will not directly establish treatment recommendations or population prevalence.

Figure 1 on the cover shows the project logic from resistance question through genetic comparison to biological interpretation.

The beneficiaries include students learning biological data analysis, researchers developing surveillance hypotheses, and public health teams interested in the genomic context of resistance. The intended contribution is a transparent, reproducible comparison of a defined genome collection, not a clinical diagnostic service.

Scope and evidence boundaries

The study uses public bacterial sequence data and associated non-identifying metadata. It excludes culturing, patient recruitment, clinical testing and laboratory manipulation. No genomes have been downloaded or analysed for this handbook. All teaching datasets are synthetic. The word evolution refers primarily to ancestry and plausible evolutionary processes; historical increases require appropriately dated, representative data and additional analysis. Remember: a genomic pattern generates an interpretation whose strength depends on the data and competing explanations.

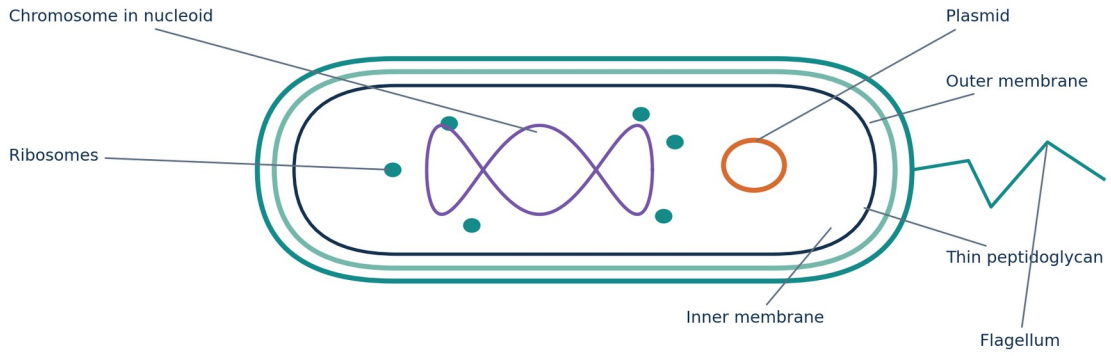
Chapter 2 Foundations of microbiology

Bacteria and prokaryotic cells

Bacteria are cellular organisms whose genetic material is not enclosed in a membrane-bound nucleus. Such cells are called prokaryotic. They possess membranes, ribosomes, metabolic systems and DNA, so they are living cells rather than simplified viruses. E. coli is a rod-shaped Gram-negative bacterium. Gram-negative describes an envelope with an inner membrane, a thin peptidoglycan cell wall and an outer membrane containing lipopolysaccharide.

The envelope matters to resistance because an antibiotic must reach its target. The outer membrane can restrict entry, porins provide channels for some molecules, and transport systems can export compounds. The periplasm is the compartment between the inner and outer membranes; beta-lactamases can act there. Calling the entire envelope a single cell wall hides these important distinctions.

A representative E. coli cell and the structures relevant to resistance



Envelope layers are expanded for clarity. The space between membranes is the periplasm.

Figure 2 Bacterial cell organisation

Legend and interpretation. Purple is chromosomal DNA; orange is a plasmid; teal dots are ribosomes. Membranes control access, ribosomes produce proteins and DNA carries inherited information.

Interpretation limit. Schematic, not to scale; plasmids and flagella are not universal in every E. coli isolate.

The chromosome occupies a nucleoid rather than a nucleus. Ribosomes translate messenger RNA into proteins. Plasmids are independently replicating DNA molecules that may carry resistance genes, but some cells have no plasmids. Flagella support motility in many strains; fimbriae support attachment. Their presence and expression vary, and neither structure should be assumed to be universal from this representative diagram.

Binary fission and inheritance

Binary fission is a form of cell division in which a bacterium duplicates its genetic material, separates the copies and divides into two descendants. Most chromosomal information passes from parent to descendants through vertical inheritance. Replication is accurate but not perfect, so descendants need not be genetically identical. Plasmids must also be maintained and partitioned; loss can occur.

A lineage expands when its descendants reproduce. In a favourable environment, a resistant lineage may increase after antibiotic exposure. Growth rate depends on nutrients, temperature, competition and other conditions, so a universal doubling time should not be assumed for natural populations.

Species strains and isolates

A species is a taxonomic grouping; bacteria are not classified by the ability to interbreed in the same way as many animals. A strain is a distinguishable genetic lineage or variant within a species. An isolate is material recovered from a particular sample or sampling event. Two isolates may belong to the same strain, and duplicate genome records may describe the same isolate. In our project, the basic observational unit should be a unique isolate represented by an assembly, with strain labels treated as metadata rather than proof of independence.

E. coli commonly inhabits intestinal environments. Many strains are commensals, while others possess combinations of traits associated with intestinal or extraintestinal disease. Resistance and virulence are different properties: a resistant isolate is not necessarily more virulent, and a pathogenic isolate need not be highly resistant. Mutations, recombination and accessory-gene acquisition generate diversity within the species. This makes E. coli useful for linking shared ancestry with variable resistance content (Tenaillon et al., 2010).

Chapter 3 DNA genes and bacterial genomes

Reading the molecular alphabet

DNA is a polymer made from nucleotides. Each nucleotide contains a sugar, phosphate and one base: adenine A, thymine T, cytosine C or guanine G. In double-stranded DNA, A pairs with T and C pairs with G. The strands are antiparallel: their chemical directions run opposite ways. Sequence files conventionally report a strand in the 5 prime to 3 prime direction.

A sequence is the ordered string of bases, not merely its length or composition. Like source code, changing one symbol can have different consequences depending on context. Unlike software, biological sequence interpretation also depends on regulation, cellular conditions, evolutionary history and molecular interactions.

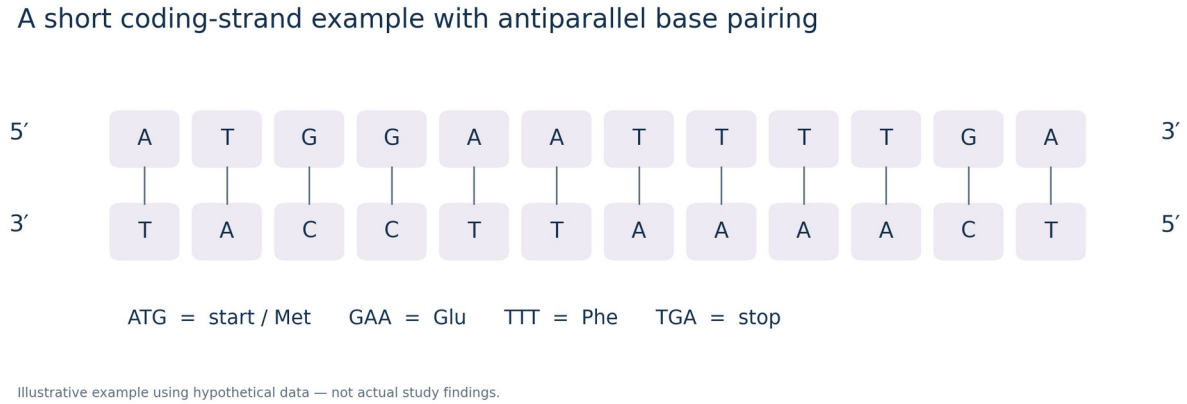


Figure 3 DNA bases and complementary strands

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Letters are bases; vertical connectors mark complementary A–T and C–G pairs. Read the upper coding strand 5′ to 3′ in the chosen three-base frame.

Interpretation limit. The sequence is hypothetical, very short and unrelated to a validated ARG.

In the explicitly hypothetical coding sequence 5′ ATG GAA TTT TGA 3′, A, T and G are nucleotides; ATG is a three-base codon. Under the standard bacterial code and this chosen reading frame, it gives methionine, glutamate, phenylalanine and a stop signal. The complementary strand is 3′ TAC CTT AAA ACT 5′. Its reverse complement, written conventionally 5′ to 3′, is 5′ TCA AAA TTC CAT 3′. This short teaching fragment is not an actual resistance gene.

Genes expression and function

A gene is a genomic region that contributes to a functional product, often a protein or functional RNA. A protein-coding gene is transcribed into messenger RNA and translated by ribosomes into an amino-acid sequence. The protein folds and acts in the cell. A beta-lactamase gene, for example, may encode an enzyme that inactivates susceptible beta-lactam antibiotics.

Promoters help regulate transcription. Coding sequences encode protein information; non-coding DNA can include promoters, terminators and functional RNA genes. Non-coding does not mean useless. A promoter mutation can increase expression of a resistance mechanism without changing the enzyme sequence.

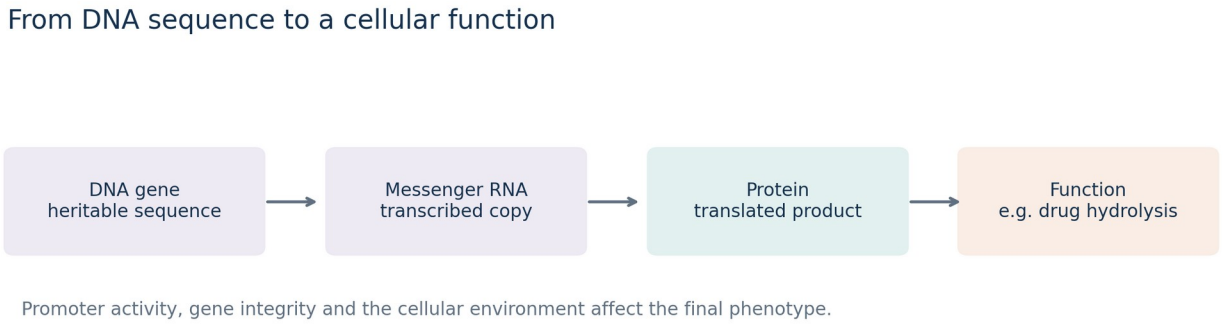


Figure 4 Gene expression and biological function

Legend and interpretation. Purple marks nucleic acids; teal marks a protein; orange marks its possible activity. Transcription and translation connect genotype with a molecular mechanism.

Interpretation limit. Some genes encode functional RNA, and protein production does not guarantee clinical resistance.

Chromosomes plasmids and genome organisation

A genome is the full genetic complement of an organism. E. coli usually has one main circular chromosome and may have plasmids. A public assembly can contain a chromosome plus plasmids or many unclosed contigs. The entire FASTA file should therefore be analysed, not just the longest contig. Plasmid location cannot reliably be inferred from the presence of an ARG on a short contig alone.

Genotype is genetic constitution; phenotype is an observable property under specified conditions. Gene presence can support a resistance prediction, but expression, functionality, interacting mechanisms and testing conditions affect the phenotype. A disrupted gene may not yield a functioning protein. Conversely, an unrecognised mechanism can produce resistance without a database hit.

Whole-genome sequencing measures many DNA fragments. Assembly attempts to reconstruct their original arrangement. Short reads may resolve individual bases well while struggling with repeated regions; long reads can span repeats and help resolve plasmids. Sequencing, assembly and annotation are different stages: measuring fragments, reconstructing sequence and assigning biological features. Remember these distinctions when reading a methods section.

Chapter 4 Antibiotics and their mechanisms

Targets and susceptibility

An antibiotic is a substance used to inhibit or kill bacteria. Different classes interfere with different cellular processes. An antibiotic must reach a biologically relevant concentration at its target; an intact target alone does not guarantee

susceptibility. Bactericidal and bacteriostatic describe effects under specified conditions and should not be treated as immutable properties in every organism and setting.

Susceptibility is evaluated with standardised testing, such as minimum inhibitory concentration measurement, and interpreted using a specified breakpoint framework and version. The minimum inhibitory concentration is the lowest tested concentration preventing visible growth under the test conditions. It is not a direct measure of how many resistance genes a bacterium carries. This handbook does not supply clinical treatment or breakpoint guidance.

Table 1 Antibiotic targets and resistance mechanisms

Antibiotic class	Target and action	Resistance route	Example determinant
Beta-lactams	Penicillin-binding proteins; inhibit peptidoglycan cross-linking	Hydrolysis; permeability and target effects	blaTEM; blaCTX-M
Tetracyclines	30S ribosomal subunit; interfere with translation	Efflux; ribosomal protection	tet(A) efflux
Fluoroquinolones	DNA gyrase and topoisomerase IV; disrupt DNA processes	Target substitutions; target protection; efflux	gyrA/parC variants; qnr
Aminoglycosides	30S subunit; disrupt accurate protein synthesis	Drug modification; target changes	aac, aph or ant families
Sulfonamides	Dihydropteroate synthase; inhibit folate synthesis	Alternative less susceptible enzyme	sul1; sul2

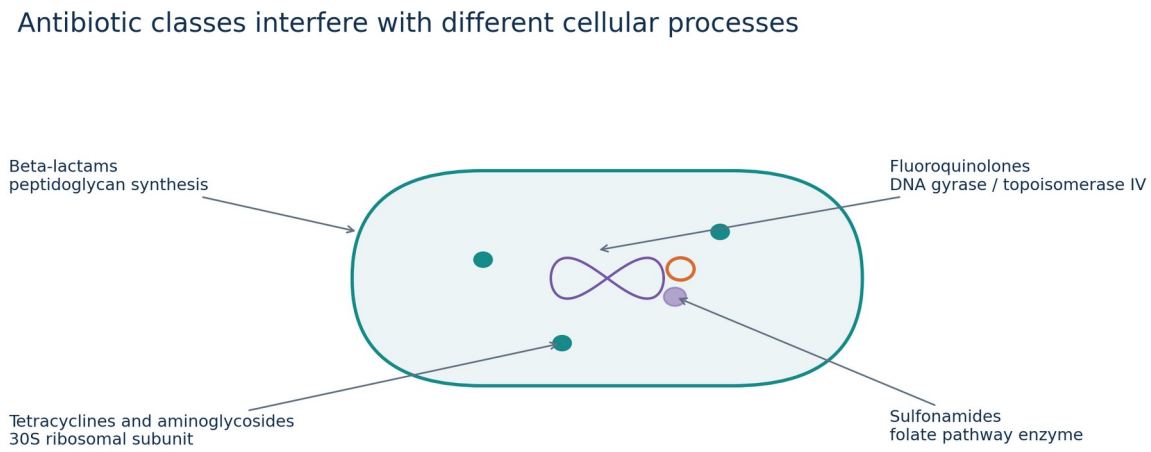


Figure 5 Antibiotic targets

Legend and interpretation. Arrows connect classes to the affected process or target location. Drugs acting at different targets can be defeated by distinct mechanisms.

Interpretation limit. Target positions are schematic; class membership alone does not determine susceptibility.

How to connect a drug target to a genomic question

Beta-lactams bind penicillin-binding proteins involved in peptidoglycan synthesis. Resistance may involve beta-lactamases, permeability changes and other mechanisms. Tetracyclines inhibit protein synthesis at the 30S ribosomal subunit; Tet(A) exports tetracyclines. Fluoroquinolones act on DNA gyrase and topoisomerase IV; resistance can involve gyrA or parC substitutions as well as acquired protection mechanisms. Aminoglycosides affect the 30S subunit, and modifying enzymes can reduce activity of particular drugs. Sulfonamides interfere with folate synthesis through dihydropteroate synthase; sul genes encode less susceptible forms of that enzyme.

These relationships guide our output structure. We should keep the determinant, mechanism and associated class in separate columns rather than replacing a gene name with a broad label such as resistant to everything. Different members of a drug class have different activity against a particular enzyme or target variant. Drug susceptibility predictions require drug-specific interpretation and validated rules, not class-level intuition (Blair et al., 2015; Munita & Arias, 2016).

Chapter 5 Antibiotic resistance mechanisms

Enzymatic inactivation and target changes

Enzymatic inactivation means a bacterial enzyme destroys or chemically modifies an antibiotic. Beta-lactamases hydrolyse the beta-lactam ring; aminoglycoside acetyltransferases, phosphotransferases and nucleotidyltransferases modify certain aminoglycosides. The relevant substrate spectrum is specific to the enzyme variant. A blaTEM family label, for instance, does not establish extended-spectrum beta-lactamase activity, because TEM-1 and extended-spectrum TEM variants differ functionally.

Target modification changes the antibiotic-binding site or its accessibility. Substitutions in the quinolone-resistance-determining regions of GyrA and ParC can reduce fluoroquinolone effectiveness. Target protection is related but distinct: Qnr proteins protect topoisomerases rather than being identical to a gyrA mutation. Recognising the difference helps explain why isolates with different genomic routes can show partially overlapping resistance phenotypes.

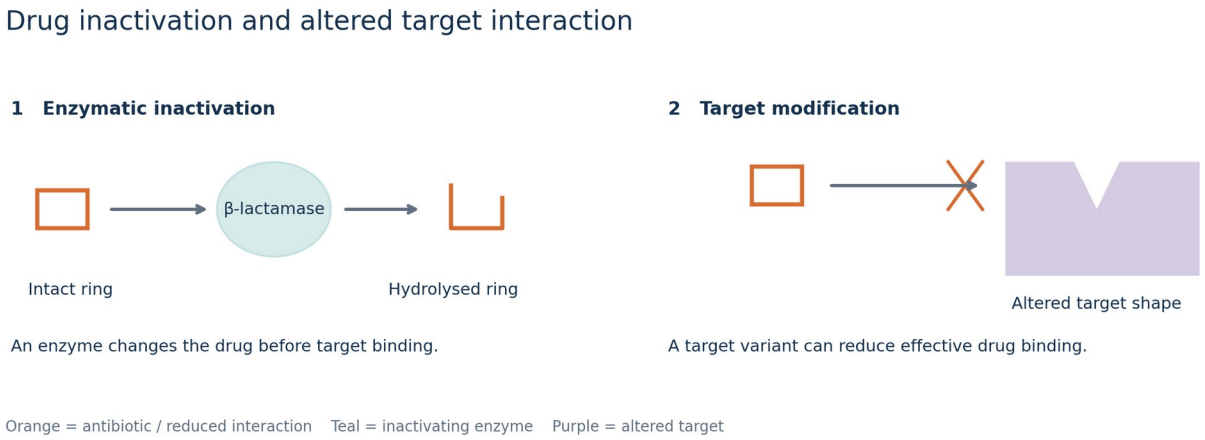


Figure 6 Enzyme and target mechanisms

Legend and interpretation. Orange outlines represent drug molecules; teal represents an inactivating enzyme; purple represents an altered target. The cross marks reduced interaction. Hydrolysis changes the drug; modification/protection changes its effective interaction with the target.

Interpretation limit. Specific enzymes and substitutions affect particular drugs; the mechanisms are not interchangeable.

Efflux permeability and alternative enzymes

Efflux pumps move substances out of the cell. Tet(A) exports tetracycline; the chromosomal AcrAB-TolC system can contribute to multidrug efflux in E. coli. A pump gene may be common across both susceptible and resistant isolates, so regulatory changes and expression often matter more than its simple presence.

Reduced permeability limits entry. Changes affecting outer-membrane porins such as OmpF or OmpC can reduce entry of some antibiotics. The interaction between reduced entry and an inactivating enzyme can be more consequential than either alone. An annotation stating porin present does not establish normal permeability.

Metabolic bypass or target replacement preserves an essential pathway despite drug inhibition. sul1 and sul2 encode alternative dihydropteroate synthases that are less susceptible to sulfonamides. Acquired dfr genes can

encode less susceptible dihydrofolate reductases associated with trimethoprim resistance. These are different targets within folate metabolism, so sulfonamides and trimethoprim should not be merged into one mechanism.

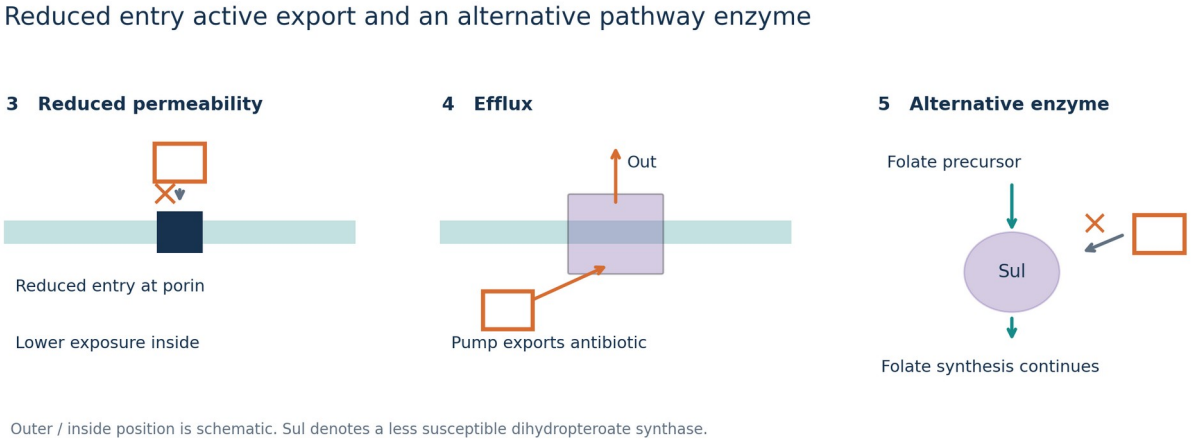


Figure 7 Permeability efflux and alternative targets

Legend and interpretation. Teal bands represent a membrane; orange squares are drug molecules; purple marks an efflux pump or alternative Sul enzyme. Arrows show movement or metabolic flow. Reducing exposure and replacing an inhibited enzyme can preserve cellular function.

Interpretation limit. Gene presence does not establish expression, dosage or the size of the phenotypic effect.

Intrinsic acquired and multidrug resistance

Intrinsic resistance arises from a species' usual structural or functional characteristics. Acquired resistance results from changes such as mutation or acquisition of foreign DNA. An acquired determinant can later be inherited vertically, so acquired describes origin rather than requiring continuous transfer.

Multidrug resistance is normally defined using phenotypic non-susceptibility across specified antimicrobial categories. Counting three ARGs is not sufficient: three genes may affect one category, and one mechanism may affect several agents. Without suitable phenotype data, report multiple resistance determinants or genomic resistance potential across classes rather than declaring a clinically confirmed multidrug-resistant isolate.

Our project can detect known determinants and compare predicted mechanisms. It cannot measure expression or susceptibility directly. Remember to retain allele names, evidence quality and uncertainty rather than allowing an attractive heatmap to hide these distinctions.

Chapter 6 Evolution of antibiotic resistance

Variation precedes differential survival

Evolution is a change in inherited characteristics of populations over generations. A point mutation changes one nucleotide. An insertion adds bases; a deletion removes them. If an insertion or deletion in a coding sequence is not a multiple of three bases, it can shift the reading frame. Mutations may be neutral, harmful or beneficial in a particular environment. Synonymous changes do not alter the encoded amino acid, although they can sometimes affect expression or other processes.

Mutations do not occur because a bacterium understands which nucleotide would help it survive. Antibiotics generally select variants that already exist or arise without being directed toward the precise beneficial outcome. Stress can influence mutation rates and gene transfer, but that does not make mutation purposeful. Resistance can also arrive through an acquired gene rather than a new point mutation.

Natural selection occurs when inherited differences affect survival or reproduction in an environment. Antibiotic exposure changes that environment. A variant that survives better may contribute more descendants and increase in relative frequency. The susceptible population may shrink while resistant descendants expand; the antibiotic does not convert every susceptible cell into a resistant one.

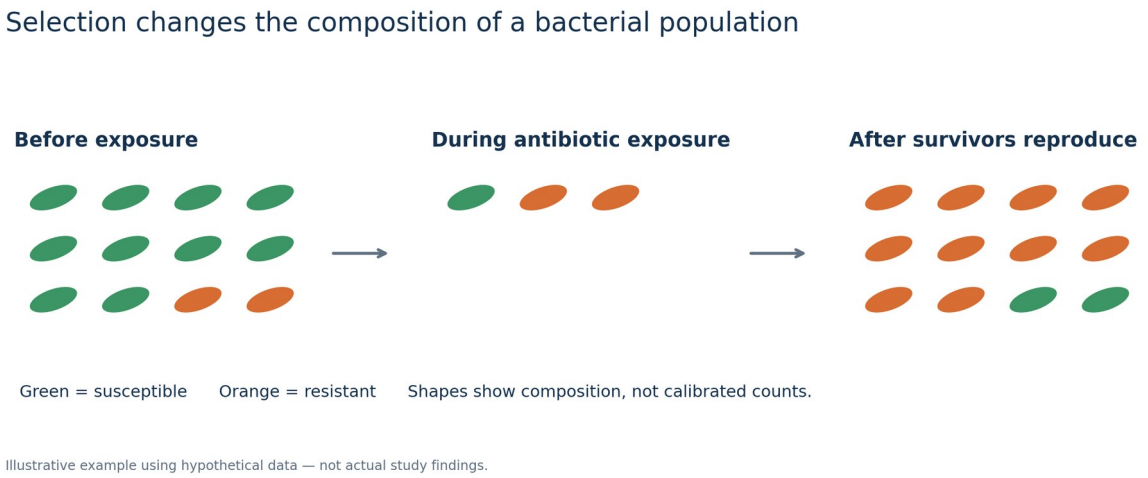


Figure 8 Natural selection under antibiotic exposure

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Green cells are susceptible; orange cells are resistant variants. Exposure favours differential survival and reproduction, changing relative frequencies.

Interpretation limit. Illustrative sequence, not a measured killing experiment or a directed mutation process.

A worked selection example

Illustrative example using hypothetical data — not actual study findings. Imagine 100 susceptible cells and 2 resistant cells. During a chosen exposure, suppose only 1 susceptible cell and both resistant cells remain viable. The resistant fraction changes from 2/102 to 2/3 before any survivor reproduces. Subsequent reproduction can further change abundance. These arbitrary counts demonstrate differential survival; they do not estimate clinical killing rates or the frequency of resistance in E. coli.

Selection is relative, whereas population size is absolute. A resistant fraction can rise even while the total number of bacteria falls. A sampling method that reports only proportions could miss this distinction. Our genome collection is not a controlled exposure experiment, so its observed frequencies cannot be interpreted as direct selection coefficients.

Fitness costs compensation and persistence

Fitness is reproductive success relative to alternatives in a specified environment. A resistance mechanism can impose costs, for example through altered enzyme performance or the burden of maintaining extra DNA. Costs vary by host background and environment. Compensation can reduce a cost while retaining resistance. Linkage to other beneficial genes and selection by other compounds can also maintain an ARG even when its associated antibiotic is absent.

Clonal expansion increases the number of descendants of a lineage. Vertical inheritance carries resistance through those descendants. Horizontal transfer introduces DNA from another cell or genetic vehicle. Gene loss removes a determinant, and recombination can replace part of a chromosome. The resulting evolutionary history can be mosaic rather than one perfectly bifurcating tree (Didelot & Maiden, 2010).

What evolution means in this study

A core-genome tree estimates relationships in the chosen genomic regions. Resistance patterns mapped onto it may be consistent with inheritance, repeated acquisition, convergence or loss. A cross-sectional collection cannot determine when a determinant first emerged or demonstrate increasing resistance over decades. Even dated samples require attention to sampling, temporal signal, recombination and model assumptions before time-calibrated inference. Remember: relatedness is evidence about ancestry, not automatically a calendar or a transmission map.

Chapter 7 Horizontal gene transfer

Conjugation

Conjugation is DNA transfer that depends on direct contact and a transfer apparatus. A conjugative plasmid can encode machinery needed for its transfer; a mobilisable plasmid requires compatible machinery supplied elsewhere. During plasmid conjugation, DNA processing and replication allow genetic material to pass while the donor usually retains a copy. A diagram showing a plasmid moving should not imply that the donor necessarily becomes plasmid-free.

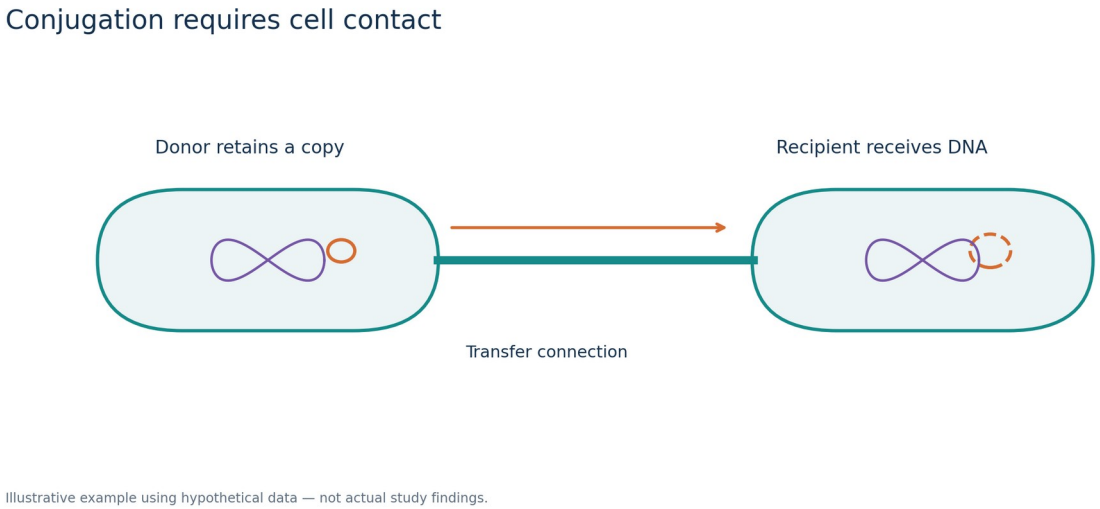


Figure 9 Conjugation requires cell contact

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Teal outlines cells; purple denotes DNA/vehicle; orange denotes transfer of genetic material. The depicted route can move genetic information if biological requirements are met.

Interpretation limit. Conceptual only; successful uptake, transfer and establishment are distinct steps. No transfer event has been measured.

The donor is the cell carrying the transferable element; the recipient receives DNA and may become a transconjugant. An ARG moves only if it is physically linked to transferred DNA and is maintained in the recipient. Compatibility, host range, restriction systems, fitness effects and ecological contact affect success. Shared ARG names alone do not demonstrate that a particular donor transmitted to a particular recipient.

Transformation

Transformation is uptake of extracellular DNA followed by stable maintenance, often through recombination. DNA can be released when cells lyse. Uptake is distinct from successful incorporation: fragments may be degraded or fail to recombine. Natural competence depends on organism and conditions. E. coli should not be presented as routinely

naturally transformable in every environment; natural transformation is a general bacterial mechanism whose contribution must be assessed rather than assumed for our collection.

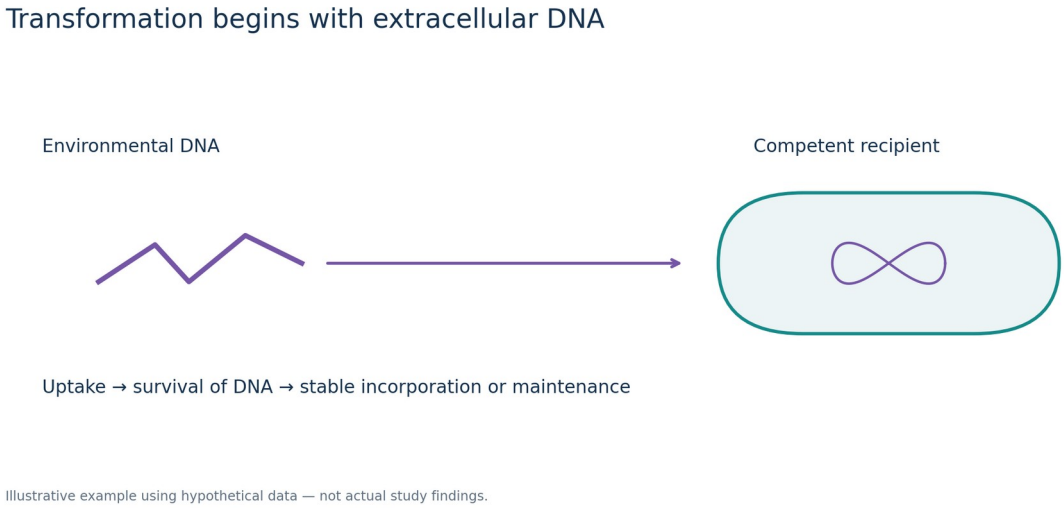


Figure 10 Transformation begins with extracellular DNA

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Teal outlines cells; purple denotes DNA/vehicle; orange denotes transfer of genetic material. The depicted route can move genetic information if biological requirements are met.

Interpretation limit. Conceptual only; successful uptake, transfer and establishment are distinct steps. No transfer event has been measured.

Transduction

Transduction is bacteriophage-mediated movement of bacterial DNA. In generalised transduction, packaging errors can place bacterial DNA in particles that transfer it to another cell. Specialised transduction involves DNA adjacent to a prophage integration site following abnormal excision. Productive transfer and stable incorporation face additional biological constraints. Detection of an ARG near a phage-like sequence is not sufficient proof of phage-mediated historical transfer.

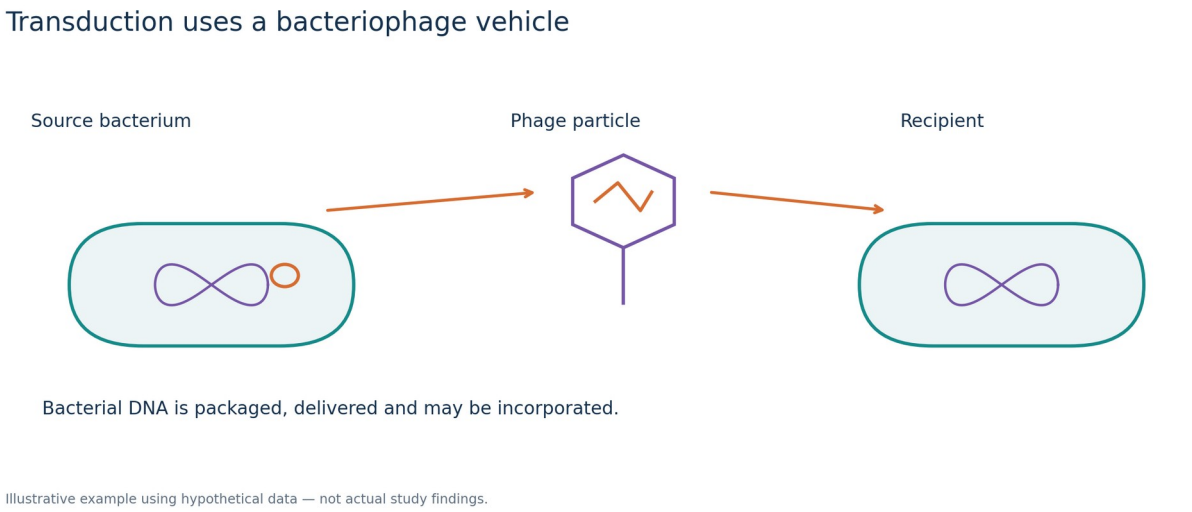


Figure 11 Transduction uses a bacteriophage vehicle

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Teal outlines cells; purple denotes DNA/vehicle; orange denotes transfer of genetic material. The depicted route can move genetic information if biological requirements are met.

Interpretation limit. Conceptual only; successful uptake, transfer and establishment are distinct steps. No transfer event has been measured.

Mobile elements are not interchangeable

A plasmid is a replicating DNA molecule. A transposon is a DNA element capable of moving between genomic locations, sometimes between chromosome and plasmid. An integron captures and expresses gene cassettes through a recombination system; it is not generally self-transmissible. Integrons can be embedded in transposons or plasmids, providing nested routes for resistance dissemination. sul1 is often associated with class 1 integron contexts, but a sul1 hit alone does not reconstruct an entire functional integron.

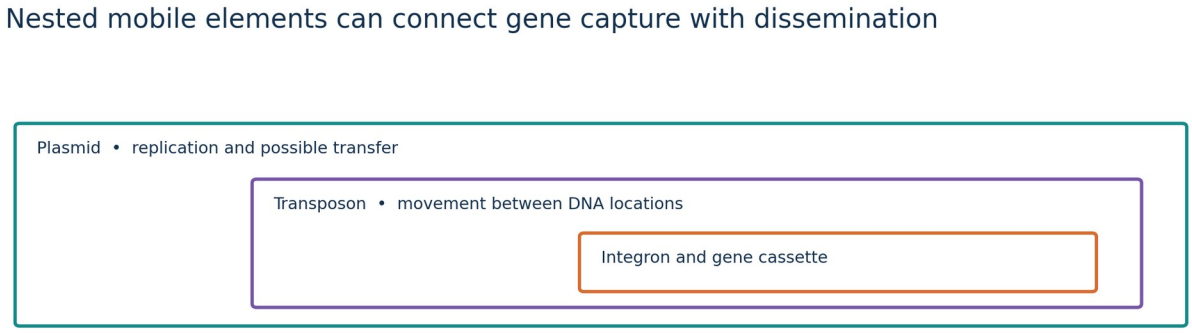


Figure 12 Plasmids transposons and integrons

Legend and interpretation. Nested outlines depict one possible physical arrangement, not every arrangement. Different elements perform replication, movement or gene-cassette capture functions.

Interpretation limit. An integron is not generally self-transmissible and can occur outside this arrangement.

For our study, HGT is a hypothesis to investigate with context. Stronger support may involve near-identical ARG alleles in otherwise divergent lineages, matching flanking regions, resolved plasmid sequences and gene-tree/species-tree discordance. These observations still require alternative explanations such as ancestral carriage, unsampled intermediates, contamination or assembly error. The appropriate conclusion is often compatible with horizontal transfer rather than confirmed transfer between these isolates.

Chapter 8 Antibiotic resistance genes

Determinants alleles and chromosomal substitutions

An antibiotic resistance gene is a gene whose product contributes to reduced antibiotic susceptibility in an appropriate biological context. A determinant is a broader term encompassing an acquired gene, a particular mutation or another causal genetic feature. An allele is a sequence variant of a gene. Family-level and allele-level identification answer different questions: blaCTX-M describes a family, whereas a specific CTX-M allele narrows sequence and functional interpretation.

Not every homolog of a resistance protein confers resistance. Database models may require sequence similarity, length coverage, conserved features or validated mutations. Exact allele calls require sufficient sequence resolution; do not rename a partial family-level match as a confirmed allele. Gene symbols and protein names should also be distinguished in writing, for example blaTEM for a gene and TEM beta-lactamase for its protein product.

Table 2 Selected resistance determinants

Determinant	Class and mechanism	Interpretation in E coli
blaTEM	Beta-lactams; enzymatic hydrolysis	Family includes variants with different spectra; TEM-1 is not automatically an ESBL.
blaCTX-M	Beta-lactams; extended-spectrum beta-lactamase activity	Resolve allele where supported; not every drug in the class has the same predicted activity.
tet(A)	Tetracyclines; active efflux	Acquired determinant; presence does not measure expression.
sul1 / sul2	Sulfonamides; alternative dihydropteroate synthase	Different genes with related effects; sul1 often occurs in class 1 integron contexts.
qnr families	Quinolones; topoisomerase protection	May contribute low-level resistance; not equivalent to confirmed high-level resistance.
aac(6′)-Ib	Certain aminoglycosides; acetylation	Substrate specificity matters; -cr is a distinct variant with additional quinolone relevance.
aph(3″)-Ib and aph(6)-Id	Streptomycin; phosphorylation	Former names strA and strB occur in literature; do not count synonyms twice.
gyrA / parC substitutions	Fluoroquinolones; altered targets	Normal gene presence is uninformative; identify validated amino-acid substitutions.

Reading the examples responsibly

TEM-1 is a narrow-spectrum beta-lactamase, whereas some TEM variants have an extended spectrum. CTX-M enzymes are associated with extended-spectrum beta-lactam activity, but drug-level predictions still depend on variant and context. Tet(A) is a tetracycline efflux determinant. sul1 and sul2 encode alternative sulfonamide targets. qnr families encode topoisomerase-protective proteins and commonly produce a contribution to quinolone resistance that should not automatically be translated into high-level clinical resistance.

Aminoglycoside resistance genes require particular care. Different aac, aph and ant enzymes modify different substrates. aac(6′)-Ib and its -cr variant are not interchangeable: the latter has additional relevance to particular fluoroquinolones. An aph(3″)-Ib/aph(6)-Id pair is associated with streptomycin resistance rather than proving resistance to every aminoglycoside.

Mutation analysis is a separate evidence layer

The chromosomal genes gyrA and parC normally perform essential topoisomerase functions. Their presence is expected and does not itself establish resistance. Known substitutions such as GyrA S83L and D87N, or ParC S80I, can be relevant to fluoroquinolone resistance; interpretation depends on the exact combination and background. The notation S83L means serine at protein position 83 is replaced by leucine, not a change at DNA nucleotide 83.

Report acquired gene calls and validated mutation calls in separate fields, then connect both to an interpretation table. Unknown nonsynonymous substitutions should remain variants of uncertain relevance unless supported by evidence. A BLAST hit to gyrA is not an adequate mutation analysis.

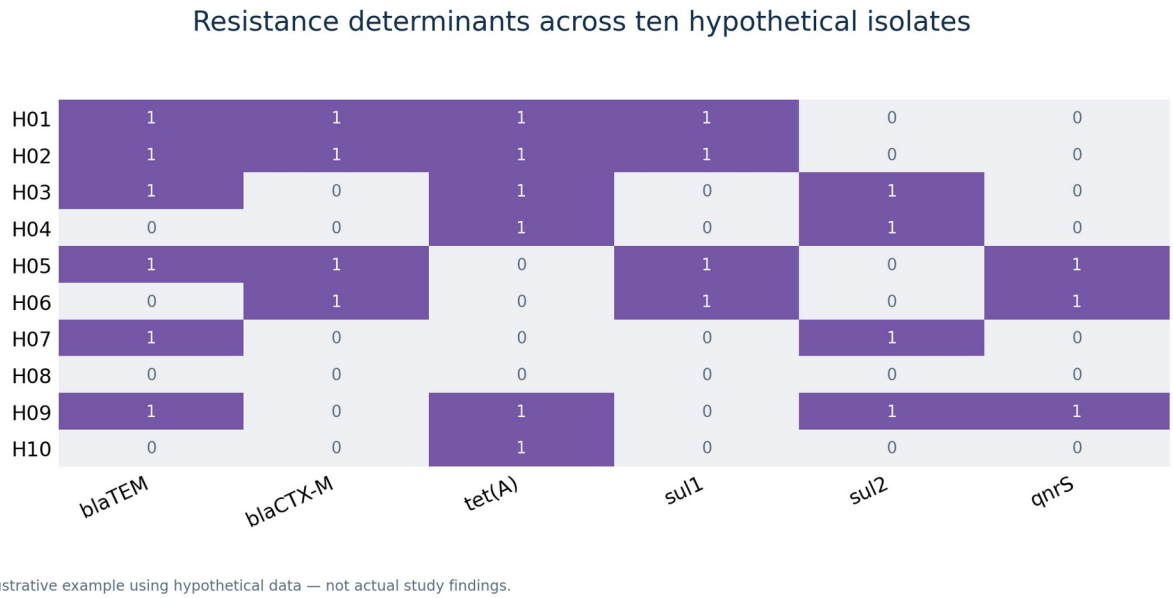


Figure 13 A conceptual resistance profile matrix

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Purple 1 = detected; pale 0 = not detected in the synthetic dataset. Profiles can differ among members of the same bacterial species.

Interpretation limit. These are hypothetical family-level calls, not measured susceptibility.

The figure uses the same synthetic isolate IDs used later in the handbook. Colour represents detection, not expression or a measured MIC. Gene count is a useful descriptive output but is not a severity scale. Remember: retaining raw calls and their evidence is essential for making an interpretable summary.

Chapter 9 Why Escherichia coli

A useful organism with important diversity

E. coli combines extensive biological knowledge with abundant public genomic data and substantial variation in its accessory genome. It inhabits human and animal hosts and can occur in environmental systems influenced by faecal material. It includes commensal and pathogenic lineages and provides opportunities to study both chromosomal variation and mobile resistance determinants. These features make it suitable for learning comparative genomics, but a broad species label does not make all isolates biologically equivalent.

The species has population structure. Closely related isolates share more ancestry than arbitrary pairs, and some lineages are repeatedly sampled because of clinical interest. If many isolates come from a single outbreak, a gene may appear frequent mainly because that clone was overrepresented. A convenience collection should not be described as an unbiased survey of the species (Tenaillon et al., 2010).

The One Health perspective

One Health recognises connections among human, animal and environmental health. Resistance determinants may be found across these settings, and genomic comparison can identify similarities that motivate investigation of shared reservoirs or exchange. It does not identify a transmission route merely because a human and an animal isolate share a gene. Shared sources, broadly distributed plasmids and incomplete sampling are alternative explanations.

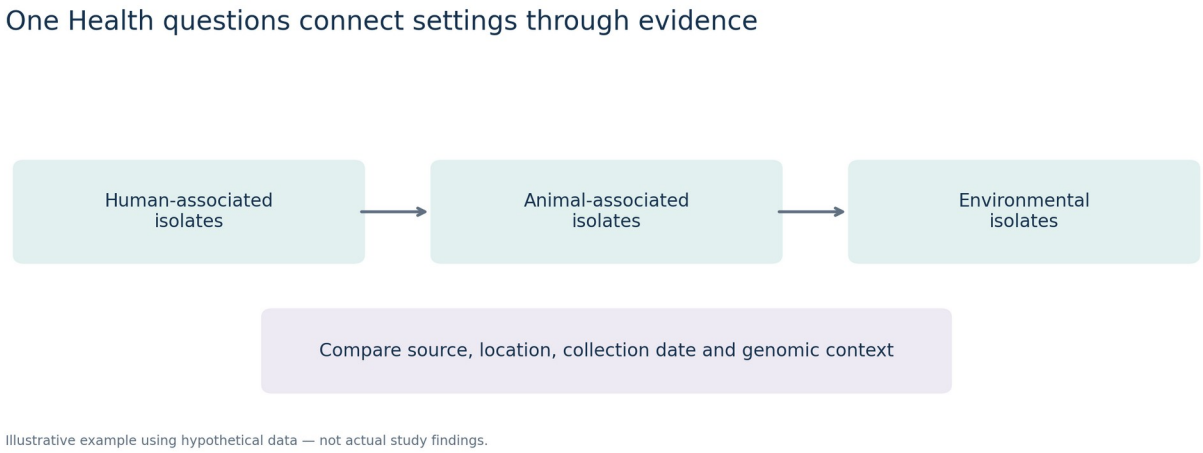


Figure 14 A One Health comparison

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Boxes represent settings; arrows identify possible connections to investigate. Metadata and genomic context help evaluate hypotheses across settings.

Interpretation limit. Arrows do not establish direction, rates or confirmed transmission.

This diagram shows connected settings and information needs rather than measured transfer rates. A project spanning settings must define how host, isolation source, geography and collection date will be obtained and harmonised. Wastewater is not a single host category, and a submission country is not necessarily the place where a sample was collected.

A manageable scope for our group

The recommended starting scope is a pilot of five genomes followed by a target collection of approximately 20–30 unique, quality-screened isolate assemblies. The number is a feasibility choice, not a power calculation or a guarantee of representativeness. Ten hypothetical isolates are used for teaching figures because a reader can inspect every row.

We should decide whether to study one reasonably coherent population or a deliberately diverse comparison. The former helps interpret fine-scale relationships; the latter highlights broader genomic variation. Mixing both without a clear question can produce confusing conclusions. Remember: choose the population before inspecting resistance results, and write down the rule.

Chapter 10 Introduction to bioinformatics

Turning sequence data into evidence

Bioinformatics uses computational methods to organise, analyse and interpret biological data. For this project the central tasks are managing genome files, identifying known resistance determinants, comparing homologous sequences and visualising patterns. A pipeline is an ordered set of transformations with defined inputs, outputs and

checks. An error early in the pipeline, such as mismatched isolate IDs, can produce plausible-looking but incorrect figures.

FASTA stores labelled DNA or protein sequences. A record begins with a greater-than sign and an identifier, followed by sequence lines. A multi-FASTA assembly may contain many contigs. FASTQ stores sequencing reads together with per-base quality scores, conventionally in four-line records. It describes measured fragments rather than a finished chromosome. A quality character is an encoded estimate, not another DNA base.

File formats and their roles

Table 3 Data formats used in the project

File type	What it contains	Use and common mistake
FASTA .fna / .faa	Labelled nucleotide / protein sequences	Genome input or protein input; do not confuse alphabet or molecule type.
FASTQ .fastq	Reads plus per-base quality	Raw sequencing; not an assembled chromosome.
GFF3 .gff	Feature coordinates and attributes	Annotation; sequence may need embedding or a linked FASTA for downstream tools.
TSV / CSV	Tabular calls or metadata	Join by stable IDs; preserve types and missingness.
Alignment .aln / .fasta	Homologous columns across samples	Tree input; gap and missing-data treatment matter.
Newick .nwk / .treefile	Tree topology and optional lengths/support	Tree exchange; a Newick file is not a sequence alignment.

A hypothetical FASTA record can be written as >DEMO_A followed by ATGGAATTTTGA. That label is a teaching identifier, not a public accession. An assembly identifier, a BioSample identifier and a read-run identifier refer to different database objects. Keep them in separate metadata fields and preserve accession versions.

Annotation locates features and assigns likely function. A coding-sequence annotation is an inference, not proof of expression or biochemical activity. Standardising annotation reduces technical differences when comparing gene content. BLAST searches for local sequence similarity. It can support homolog identification, but the highest-scoring hit is not automatically the exact resistance allele or a validated resistance prediction.

Alignment variants and comparison

Sequence alignment arranges homologous bases or amino acids into comparable columns. Gaps represent inferred insertions or deletions relative to other sequences. Aligning unrelated sequences merely to obtain an output is biologically unhelpful. Variant detection identifies differences against a reference or within an alignment; the meaning of a coordinate depends on that reference, orientation and coordinate convention.

Comparative genomics studies shared and differing genomic features. Phylogenetics reconstructs relationships under a model of sequence change. An ARG presence/absence matrix and a core-sequence alignment are different representations and should not be silently substituted for one another.

Computing habits that matter biologically

Use one stable sample identifier across filenames, tables, tree tips and plots. Validate joins as one-to-one where appropriate; a many-to-many merge can inflate counts. Preserve missing values instead of converting them to zeros. Retain original files as read-only inputs and write derived files elsewhere. A reproducible run records tool versions, database versions, parameters, checksums and logs, not only the final image. Remember: biological credibility depends on data engineering as well as algorithms.

Chapter 11 Biological databases and data acquisition

Choosing the right resource

Table 4 Resources for data and resistance analysis

Resource and URL	Data and advantage	Limitation or selection note
GenBank https://www.ncbi.nlm.nih.gov/genbank/	Submitted sequences and annotations; broad coverage	Submission quality and metadata vary.
RefSeq https://www.ncbi.nlm.nih.gov/refseq/	Reference sequence collection; organised processing	Not an independent sample from linked GenBank records.
NCBI Datasets https://www.ncbi.nlm.nih.gov/datasets/	Download interface, packages and metadata reports	Record exact search date, accession versions and filters.
SRA https://www.ncbi.nlm.nih.gov/sra	Raw reads and experiment links	Read QC/assembly increases computing work.
ENA https://www.ebi.ac.uk/ena/browser/	Raw reads, assemblies and sample records	Cross-archive duplicates require deduplication.
CARD https://card.mcmaster.ca/	Curated determinants, ontology and RGI models	Model category and database snapshot affect interpretation.
AMRFinderPlus https://github.com/ncbi/amr	Curated gene/mutation calling tool and reference data	Organism settings and input compatibility matter.
ResFinder https://genepi.food.dtu.dk/resfinder	Acquired genes and related genotype prediction resources	Record thresholds, supported species and mutation-analysis settings.

GenBank archives submitted sequences and annotations. RefSeq provides an organised reference collection with its own inclusion and annotation processes. NCBI Datasets provides interfaces and packages for retrieving sequences and metadata; it is an access service rather than a separate experimental dataset. SRA and ENA provide raw reads and linked records. GenBank, ENA and DDBJ exchange sequence data, so cross-database records are not necessarily independent samples.

CARD is a curated resistance knowledgebase with detection models; its Resistance Gene Identifier interprets sequence against those models (Alcock et al., 2023). AMRFinderPlus is an analysis tool linked to NCBI curated reference data, not simply another raw-genome repository (Feldgarden et al., 2021). ResFinder provides acquired-gene analysis and related prediction resources; mutation analysis must use the relevant supported functionality and settings (Bortolaia et al., 2020). These resources differ in nomenclature, scope and decision rules, so agreement is useful but not an unquestionable gold standard.

A prespecified selection protocol

Before downloading, define the study population, date of database search, search terms, inclusion rules and exclusion rules. Prefer assemblies with clear isolate metadata and suitable quality. For a first project, complete or

high-quality assemblies can reduce unresolved plasmid and fragmentation problems. Restricting to complete assemblies also introduces selection bias, so explain that tradeoff.

Record the versioned assembly accession, BioSample, BioProject, strain/isolate label, host, isolation source, collection date, collection location, assembly level, assembly length, contig count, platform if available, and citation. Missing metadata remain missing. A publication date must not be used as a substitute for collection date. De-duplicate the same BioSample or isolate represented by paired GenBank/RefSeq records or repeated submissions.

Quality and sampling decisions

Completeness means how much expected genome content is represented. Contiguity concerns how fragmented the assembly is. Contamination means sequence from another source is included. A large N50 can indicate longer contigs but does not establish correct sequence, correct species or complete plasmid recovery. QUAST summarises assembly properties; interpretation still requires biological and record-level checks (Gurevich et al., 2013).

As provisional screening flags, review assemblies far outside a roughly 4–6.5 Mb range or with hundreds of contigs; these are review triggers, not universal biological exclusion laws. Check taxonomic consistency, unusual duplicated markers if assessed, ambiguous bases and assembly warnings. Do not remove inconvenient resistance profiles under the guise of quality control.

Feasible download route

Use the NCBI Datasets web interface to preview and select records, export the accession list and metadata, then download genome FASTA packages for those exact accessions. The command-line alternative is shown in Chapter 14. Start with five genomes to test identifiers, outputs and storage. Archive the accepted and excluded lists with reasons. Public collections overrepresent well-resourced studies, outbreaks and resistant isolates; sample frequency describes our selected collection, not national or global prevalence.

Chapter 12 Comparative genomics

Core accessory and pangenome

The core genome is the set of homologous genes or genomic regions shared across the analysed collection under an explicit definition. The accessory genome contains variably present features. The pangenome is their union. These are dataset-dependent concepts: adding more genomes or changing clustering thresholds can change membership. A strict core requires every isolate; a soft core allows a prespecified small amount of missingness.

Orthologs generally descend from a shared ancestral gene separated by lineage divergence. Paralogs arise from duplication. Gene clustering algorithms approximate these relationships using sequence and context. Fragmented genes and inconsistent annotation can falsely inflate accessory content. Panaroo addresses important sources of error in prokaryotic pangenome analysis, but its output still depends on inputs and settings (Tonkin-Hill et al., 2020).

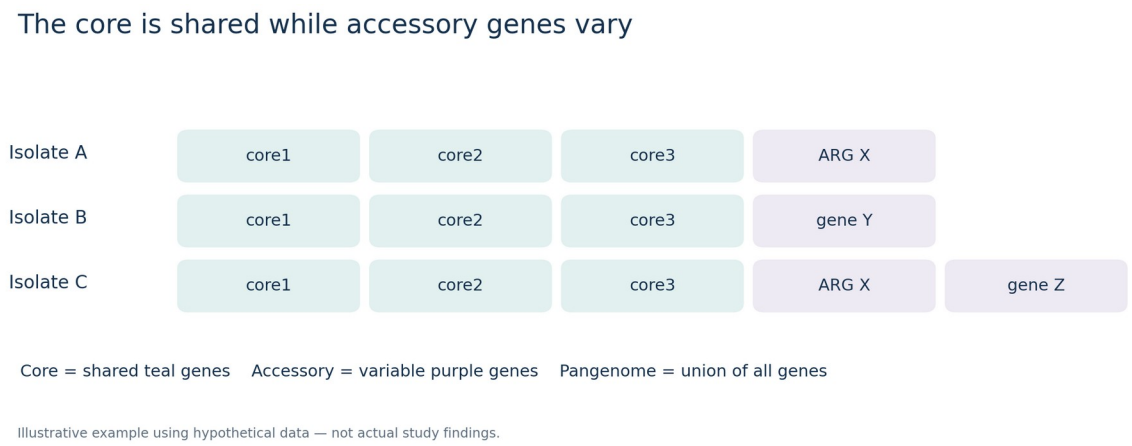


Figure 15 Core accessory and pangenome

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Teal features occur in all three hypothetical isolates; purple features vary. The union of core and accessory gene families forms the sample pangenome.

Interpretation limit. Membership depends on sample selection and gene-clustering rules.

Worked sequence comparison

Illustrative example using hypothetical data — not actual study findings. Compare A: ATGGAATTTTGA, B: ATGGAATCTTGA and C: ATGGAATTTTGA. B differs from A at nucleotide 8. In the chosen coding frame, TTT becomes TCT, changing phenylalanine to serine. The example shows a nonsynonymous SNP; it does not establish resistance, because these are invented short sequences with no validated ARG function.

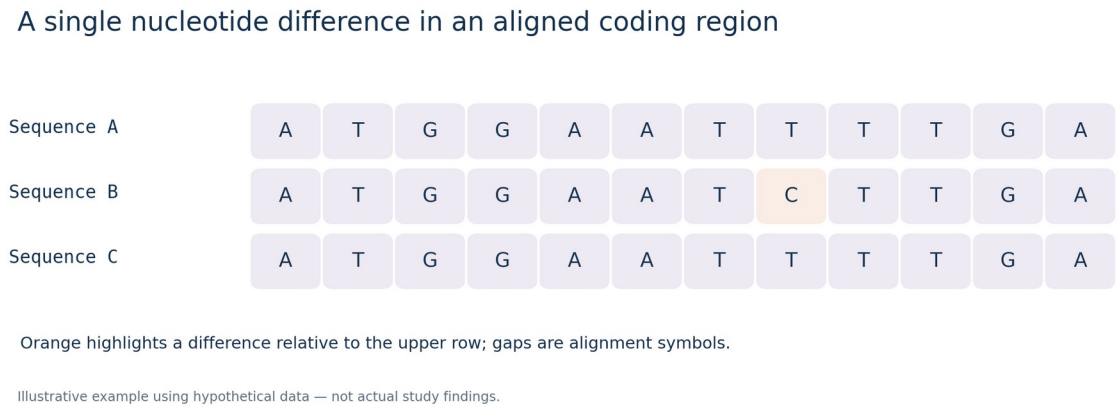


Figure 16 A single nucleotide difference in an aligned coding region

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Letters are aligned bases; orange marks differences; dash denotes an inferred gap. Aligned columns let us distinguish conserved positions from changes.

Interpretation limit. Toy sequences lack validated resistance function and are not actual genomes.

A gap indicates an inferred insertion/deletion relationship in an alignment, not necessarily a sequencing failure. Distinguish a true genomic deletion from a missing assembly segment. For real variants, record the reference accession/version, genomic coordinate, reference and alternate base, gene, strand, codon and amino-acid effect. A protein position must not be confused with an absolute genome coordinate.

Comparing resistance profiles

Construct a matrix with one row per accepted isolate and one column per prespecified gene family or allele. Use 1 for a supported call, 0 for no qualifying detection in adequate data, and NA for an unreliable or unassessable state.

Save the raw calls before collapsing multiple hits. Decide whether family and allele analyses are separate so the same determinant is not counted twice.

Gene frequency is the number of assessable isolates with a qualifying call divided by the number assessed for that determinant. If 6 of 10 teaching isolates have a gene, the displayed sample frequency is 60%; if two isolates are unassessable, the denominator must be reconsidered and reported. Counting hits rather than isolates would overcount duplicated genes.

Jaccard similarity for binary profiles is shared detected genes divided by genes detected in either isolate. Shared non-detections do not contribute. Two all-zero profiles yield a zero denominator and need an explicit convention or missing result. Profile similarity describes shared content, not automatically shared ancestry.

Chapter 13 Phylogenetics and evolutionary relationships

What a tree represents

A phylogenetic tree is a model-based representation of inferred relationships among sampled sequences. Tips represent sampled isolates or sequences; internal nodes represent inferred common ancestors; branches connect them. A clade contains an ancestor and all its descendants. Rotating branches around a node changes the drawing but not the relationships.

A rooted tree identifies an ancestral direction relative to a root; an unrooted tree represents splits without a specified ancestral direction. An outgroup requires biological justification. Midpoint rooting uses a distance-based assumption and should be labelled as a display choice when used. Left-to-right placement alone does not make a tree time-calibrated.

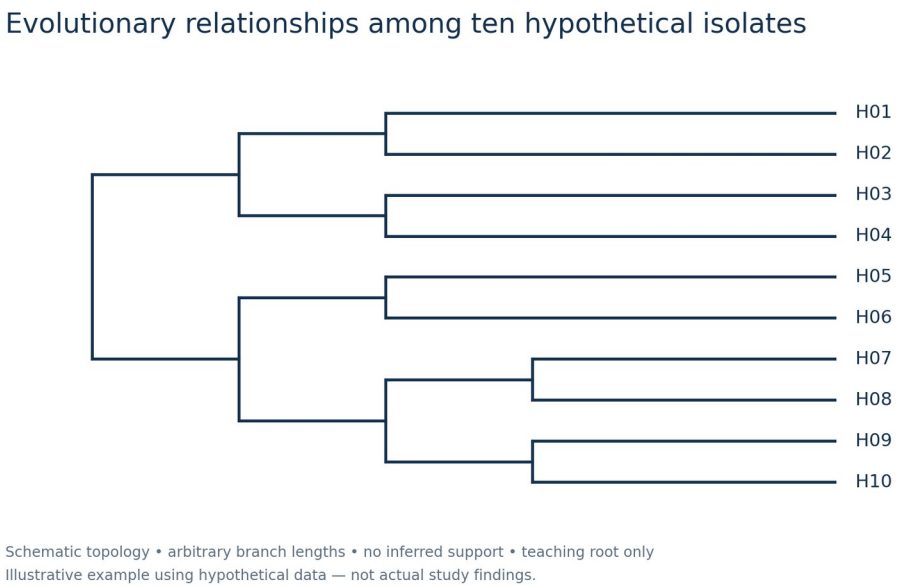


Figure 17 Evolutionary relationships among ten hypothetical isolates

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Tips are isolates; internal junctions are common ancestors; horizontal lengths are arbitrary. Compare shared nodes to assess drawn relatedness; in the combined panel read across to the matching profile.

Interpretation limit. No tree inference or transfer reconstruction was performed; branch lengths and the root are teaching conventions.

This schematic uses ten hypothetical isolates. The horizontal scale is deliberately arbitrary and branch supports are not supplied, because no alignment or inference has been run. The drawn root is a teaching convention. A real phylogram would identify branch-length units, such as substitutions per site, and rooting method. Tip order is not a ranking of age, pathogenicity or resistance.

Constructing a defensible genomic tree

For this project, use consistently annotated genomes to identify single-copy core genes, align homologous loci and concatenate them in a consistent order. Assess missingness and problematic columns. Recombination can create local histories different from the clonal history; where feasible, assess and mask likely recombinant regions using a suitable method. If not assessed, explicitly describe the tree as a core-genome relationship estimate potentially influenced by recombination.

A core-SNP tree uses variable sites from comparable core regions. It must address ascertainment bias if invariant sites are excluded; a full core alignment includes invariant sites and avoids that specific omission. Mapping-based SNP analysis also depends on reference choice and callable sites. Raw sequencing reads provide quality evidence not recoverable from a draft assembly alone.

Models and branch support

Maximum-likelihood inference seeks a tree and model parameters that make the observed alignment comparatively likely. IQ-TREE provides model selection, tree search and support estimation (Minh et al., 2020). Bootstrap methods assess stability under resampling; a 95 support label is not a 95% probability that an isolate directly infected another. Ultrafast and conventional bootstrap values are not identical quantities.

Short internal branches, recombination, poor alignment and limited informative sites can weaken relationships. High support does not rescue biased sampling or an incorrect model. Identical core sequences can occur in distinct isolates; unresolved branching should not be disguised as confidently ordered transmission.

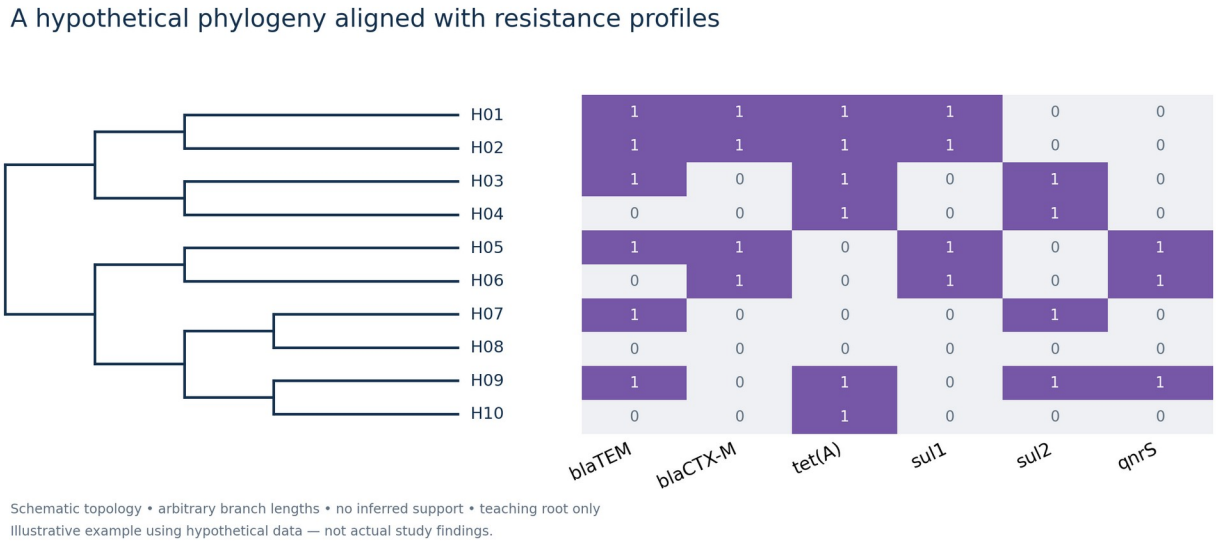


Figure 18 A hypothetical phylogeny aligned with resistance profiles

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Branches represent hypothetical ancestry; rows align with H01–H10; purple cells denote synthetic detections. Compare shared nodes to assess drawn relatedness; in the combined panel read across to the matching profile.

Interpretation limit. No tree inference or transfer reconstruction was performed; branch lengths and the root are teaching conventions.

Mapping resistance onto a tree

Annotating the same tips with a resistance matrix lets us ask whether content tracks ancestry. Shared genes in a clade may reflect inheritance or a shared exposure/acquisition event; patchy occurrence can reflect acquisition, loss, convergent changes or technical failure. Constructing a tree from ARG profiles and then claiming that ARGs cluster by that tree would be circular. The evolutionary tree should use independent core sequence evidence. Remember: the tree and heatmap answer complementary questions.

Chapter 14 Proposed project methodology

Design and practical resources

This is a retrospective comparative genomic study using public isolate assemblies. A target of 20–30 accepted genomes, after a five-genome pilot, balances interpretability and student resources. It is not a formal power calculation. Use an ordinary laptop with approximately 8–16 GB RAM for small batches where feasible; actual memory, disk and runtime depend on genome count, annotation databases and tool settings. Run heavy steps sequentially, limit threads and use institutional compute if a pilot exceeds capacity. On Windows, WSL2 or a supported container environment is more practical than assuming all bioinformatics tools run natively.

Keep the minimum deliverable focused: accession/metadata manifest, quality report, acquired-gene and known-mutation calls, profile matrix, core-genome tree, combined plot and reproducible interpretation. Raw-read assembly, large metagenomic analyses, whole-species association studies and time-calibrated transmission inference are extensions, not required baseline tasks.

Table 5 Stage inputs outputs and purpose

Stage	Input and tool option	Output and purpose
1 Literature review	Research question; primary papers and official methods	Evidence table and defined scope.
2 Dataset identification	NCBI Datasets metadata search	Candidate accession list and eligibility rules.
3 Selection	Metadata, duplicate checks, collection definition	Accepted/excluded manifest with reasons.
4 Download	Verified accession list; Datasets web/CLI	Genome FASTA, reports and checksums.
5 Quality assessment	FASTA and records; QUAST plus record/taxonomy review	QC report and final accepted manifest.
6 Annotation	Accepted FASTA; consistent Bakta or compatible alternative	GFF and proteins for comparable gene features.
7 Resistance detection	Accepted assemblies; AMRFinderPlus, targeted cross-check	Raw gene/mutation calls with evidence.
8 Comparison	Calls and annotations; scripts and optional Panaroo	Profiles, frequencies and comparable core features.
9 Phylogeny	Core alignment; IQ-TREE with documented model/support	Tree, log, model and uncertainty assessment.
10 Profile comparison	Tree plus matrix with identical IDs	Descriptive lineage/profile comparison.
11 Visualisation	Validated tables and tree; plotting software	Charts, heatmap and combined panel.
12 Interpretation	Outputs, context and literature	Evidence-linked biological explanations.
13 Conclusions	Methods, limitations and objective mapping	Bounded conclusions and reproducibility archive.

Stepwise protocol and decision gates

First, agree the population and record a search protocol. Conduct a literature review separating molecular mechanism papers, E. coli population studies and computational validation papers. Screen candidate records and

retain a rejection log. Download exact accepted assemblies and metadata; compute file checksums so later changes can be recognised. Confirm that sample identifiers are unique and that all intended replicons are included.

Second, run assembly assessment and review taxonomic and metadata warnings. Quarantine suspicious files until resolved; never silently drop them from only one analysis. If too few genomes remain, revise scope transparently before inspecting resistance patterns. The accepted manifest is the common input to all later steps.

Third, standardise annotation for gene-content/core analysis, using a compatible pipeline and database snapshot. Bakta is one option (Schwengers et al., 2021); verify database size and downstream GFF compatibility during the pilot. AMRFinderPlus can analyse assembled nucleotide sequence independently, so consistent annotation is particularly important for the pangenome branch rather than a prerequisite for every nucleotide-based resistance call.

Fourth, run AMRFinderPlus with the supported Escherichia organism setting, retaining acquired genes and validated mutation results as distinct classes. Record software/database versions, full command and raw output. Keep the tool's curated evidence rules as the baseline rather than lowering thresholds to create more positives. Review partial, interrupted, low-coverage or ambiguous calls separately. Use CARD/RGI or ResFinder as a targeted sensitivity comparison on discordant or selected calls, not an excuse to mix incompatible call types.

Practical command templates

These commands are starting templates for a prepared Linux/WSL environment, not evidence that analysis has been performed. Install and validate each tool using its own documentation, check the installed executable name and pin the environment before the full run. accession_list.txt must contain the group's verified, versioned assembly identifiers; none are invented here.

```
datasets download genome accession --inputfile accession_list.txt --include genome,seq-report --filename genomes.zip
unzip genomes.zip -d data_download
amrfinder -u
amrfinder -V > amrfinder_version.txt
amrfinder -l > supported_organisms.txt
amrfinder -n genomes/ID.fna -O Escherichia --threads 2 -o amr/ID.tsv
quast.py genomes/*.fna -o qc/quast -t 2
```

Update the resistance database once during setup, then freeze the snapshot for the study; updating between isolates can make calls incomparable. -O Escherichia enables supported organism-specific mutation screening. A nucleotide input must be an assembled sequence, not raw FASTQ reads. The exact output schema and available calls may evolve, so preserve headers rather than parsing arbitrary column numbers (National Center for Biotechnology Information, n.d.-b).

Core alignment and inference templates

After annotation, confirm that one GFF with the required associated sequence is available per accepted isolate and that your installed Panaroo version accepts the chosen annotation output. The following illustrates the core-alignment route; filenames should be checked against actual outputs. Strict cleaning may discard real rare genes, so review warnings and compare an appropriate sensitivity setting if important determinants appear affected.

```
panaroo -i annotations/*.gff -o core --clean-mode strict -a core --aligner mafft -t 2
iqtree2 -s core/core_gene_alignment.aln -m MFP -B 1000 -T 2 -seed 2026 --prefix trees/core
```

The IQ-TREE template assumes a full nucleotide core alignment including invariant sites. If using SNP-only data, use a valid ascertainment-correction approach and its required assumptions; do not apply the same model blindly. The seed supports repeatability, not biological accuracy. Read the log and report the selected model, alignment length, informative sites, support method, root/display convention and recombination treatment.

Data model and quality checks

Create an isolate table keyed by sample_id, a raw-call table with one row per call and a derived matrix keyed by sample_id and determinant. Raw calls should retain contig/coordinates, allele or symbol, match method, percent identity, coverage and available completeness flags. Not every field is supplied by every tool, so preserve the native schema and document transformations.

Before plotting, verify accepted isolate counts, unique IDs, coordinate consistency and the match between matrix rows and tree tips. For each determinant, calculate n_detected, n_assessable and n_detected/n_assessable. Inspect at least one positive, one negative and every ambiguous case against raw files. Verify that a zero means an assessable non-detection rather than a missing file.

Analysis interpretation and reproducibility

Compare within-clade and between-clade patterns descriptively. Small, structured samples do not justify treating all isolate pairs as independent observations in a simple significance test. Optional association work requires lineage-aware methods, sufficient data, confounder assessment and multiple-testing control; it is beyond the baseline scope.

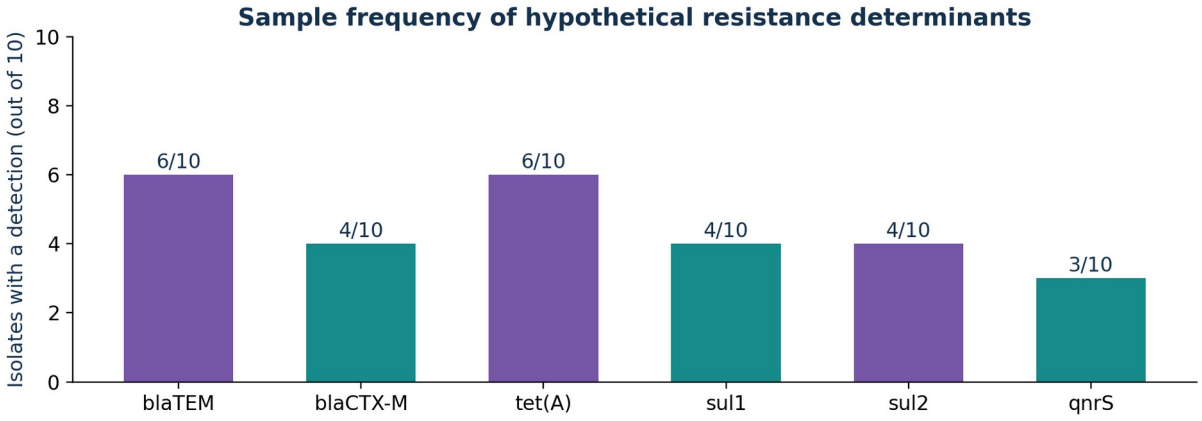
Organise folders for metadata, immutable inputs, quality reports, annotations, raw AMR calls, core alignments, trees, figures and scripts. Save environment locks or container digests, commands, logs, checksums, database release dates and a README. Suggested responsibilities are data curation, resistance analysis, phylogeny, and integration/interpretation, with cross-review of outputs. A six-week plan can allocate weeks 1–2 to reading and selection, 3 to QC and annotation, 4 to AMR/core analysis, 5 to figures and interpretation and 6 to review and reporting. This is a proposed schedule, not completed progress.

Chapter 15 Expected findings and visualisations

One teaching dataset and seven output types

Illustrative example using hypothetical data — not actual study findings. Every chart in this chapter uses invented isolate IDs H01–H10 and six determinant columns. A value of 1 means detected within the teaching example; a 0 means not detected within that complete synthetic matrix. There are no uncertain entries here solely to keep the example readable. Real project matrices should preserve NA where appropriate.

The columns are blaTEM, blaCTX-M, tet(A), sul1, sul2 and qnrS. They are grouped at family level for teaching and do not establish particular alleles or MIC values. The synthetic matrix and figure metadata are included with the separate diagram collection, allowing readers to reproduce the counts. The schematic tree was drawn independently for teaching rather than inferred from these data.



Illustrative example using hypothetical data — not actual study findings.

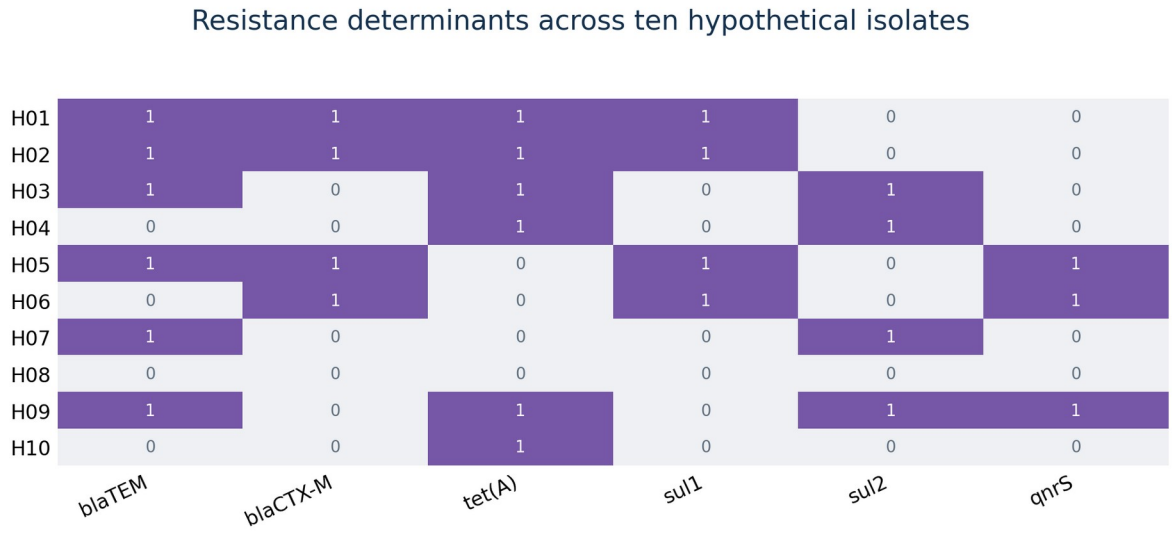
Figure 19 Figure A Resistance gene frequency

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Bar height counts unique synthetic isolates carrying each determinant. Counts are 6, 4, 6, 4, 4 and 3 out of ten in displayed column order.

Interpretation limit. Sample frequencies are not population prevalence or actual results.

Figure A is a resistance gene frequency bar chart. For each determinant, the bar height is the number of synthetic isolates with a 1; labels give counts out of ten. The denominator is isolate count, not the sum of gene calls. A bar can be high because a particular clone was oversampled. The chart helps identify common determinants in the chosen collection but cannot estimate population prevalence, temporal trends or clinical severity.



Illustrative example using hypothetical data — not actual study findings.

Figure 20 Figure B Resistance gene presence and absence

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Purple 1 = detected; pale 0 = not detected in the synthetic dataset. Profiles can differ among members of the same bacterial species.

Interpretation limit. These are hypothetical family-level calls, not measured susceptibility.

Figure B is an antibiotic resistance gene heatmap. Rows are isolates; columns are determinant families. Purple cells with 1 indicate synthetic detections and pale cells with 0 indicate synthetic non-detections. The legend is categorical

rather than a concentration scale. Multiple filled cells show co-occurrence in the same isolate, not physical linkage on one plasmid. The matrix can reveal candidate profile groups but does not itself reconstruct ancestry.

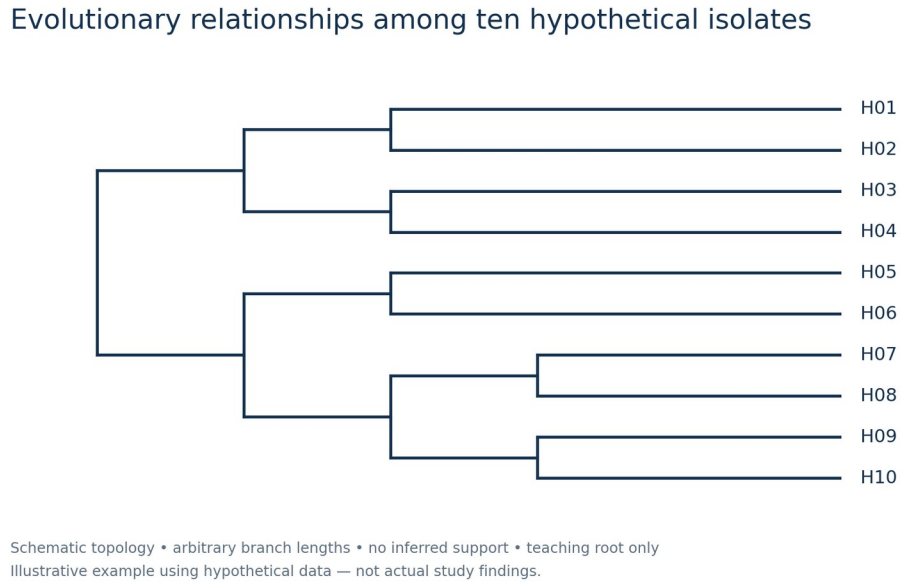


Figure 21 Evolutionary relationships among ten hypothetical isolates

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Tips are isolates; internal junctions are common ancestors; horizontal lengths are arbitrary. Compare shared nodes to assess drawn relatedness; in the combined panel read across to the matching profile.

Interpretation limit. No tree inference or transfer reconstruction was performed; branch lengths and the root are teaching conventions.

Figure C is a phylogenetic tree display using a hypothetical topology. Sister tips share a more recent drawn node than they do with tips across deeper splits. All branch lengths are arbitrary and no statistical supports were calculated. In actual work this panel would be inferred from a quality-controlled core alignment and would include the scale, model and support details. The drawing teaches tree reading; it is not a computed biological result.

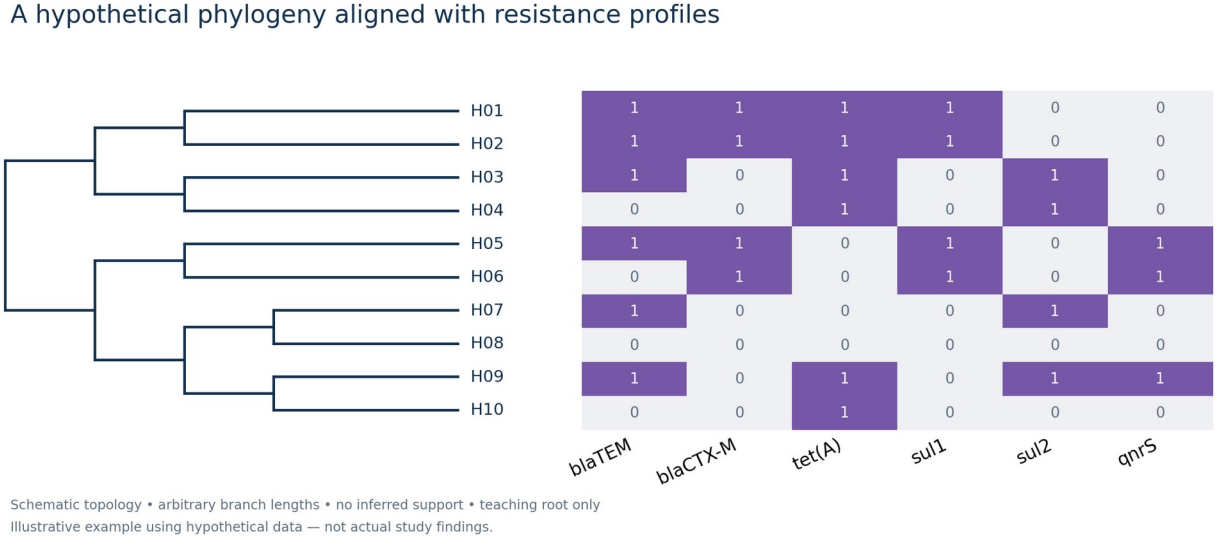


Figure 22 A hypothetical phylogeny aligned with resistance profiles

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Branches represent hypothetical ancestry; rows align with H01–H10; purple cells denote synthetic detections. Compare shared nodes to assess drawn relatedness; in the combined panel read across to the matching profile.

Interpretation limit. No tree inference or transfer reconstruction was performed; branch lengths and the root are teaching conventions.

Figure D aligns a tree with the same resistance matrix. Tip order and heatmap rows match exactly, so patterns can be read across the panels. For example, neighbouring H01–H02 share several synthetic determinants, while a determinant appearing elsewhere might prompt an acquisition/loss hypothesis. The topology does not demonstrate donor direction or transfer events. Row alignment is a substantive correctness check, not merely a formatting choice.

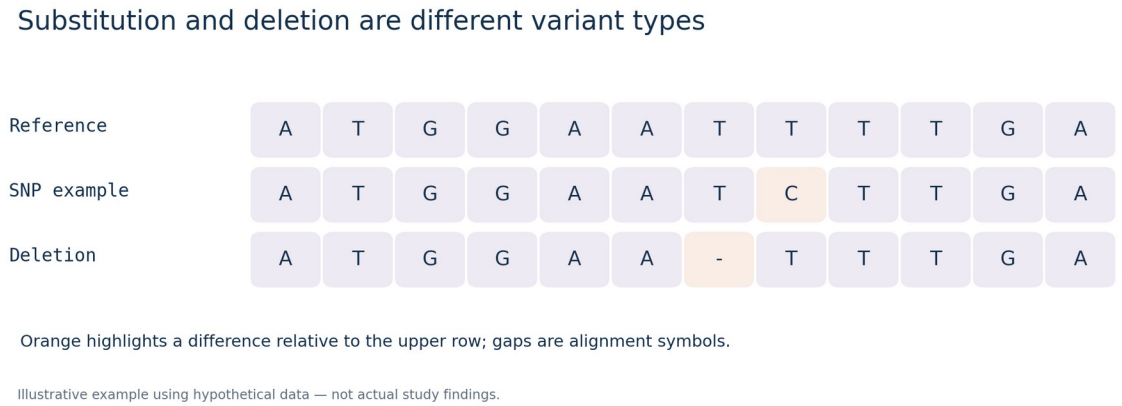


Figure 23 Substitution and deletion are different variant types

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Letters are aligned bases; orange marks differences; dash denotes an inferred gap. Aligned columns let us distinguish conserved positions from changes.

Interpretation limit. Toy sequences lack validated resistance function and are not actual genomes.

Figure E shows genomic variation. A coloured base highlights a hypothetical substitution and a gap marks an inferred deletion in a separate toy alignment. Coordinates refer only to the displayed teaching region. The panel distinguishes observed sequence differences from causal resistance determinants: a real variant needs validated functional interpretation and reliable sequence support before being described as resistance-associated.

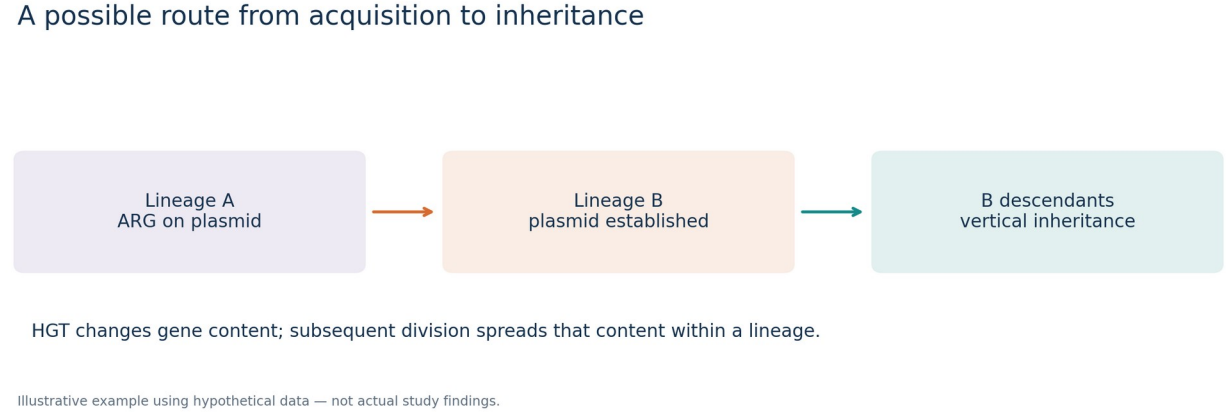


Figure 24 Figure F A conceptual resistance dissemination route

Illustrative example using hypothetical data — not actual study findings.

Legend and interpretation. Purple marks the initial gene context; orange marks acquisition; teal marks descendants. Horizontal acquisition and vertical inheritance can operate in sequence.

Interpretation limit. This is one plausible scenario and identifies no actual donor or recipient.

Figure F illustrates a plausible horizontal-transfer pathway: an ARG on transferable plasmid DNA becomes established in another lineage, then is inherited by descendants. The arrow is a conceptual scenario, not a reconstructed event between project isolates. To examine such a hypothesis, compare full plasmid or flanking sequence context and account for possible unsampled intermediates.

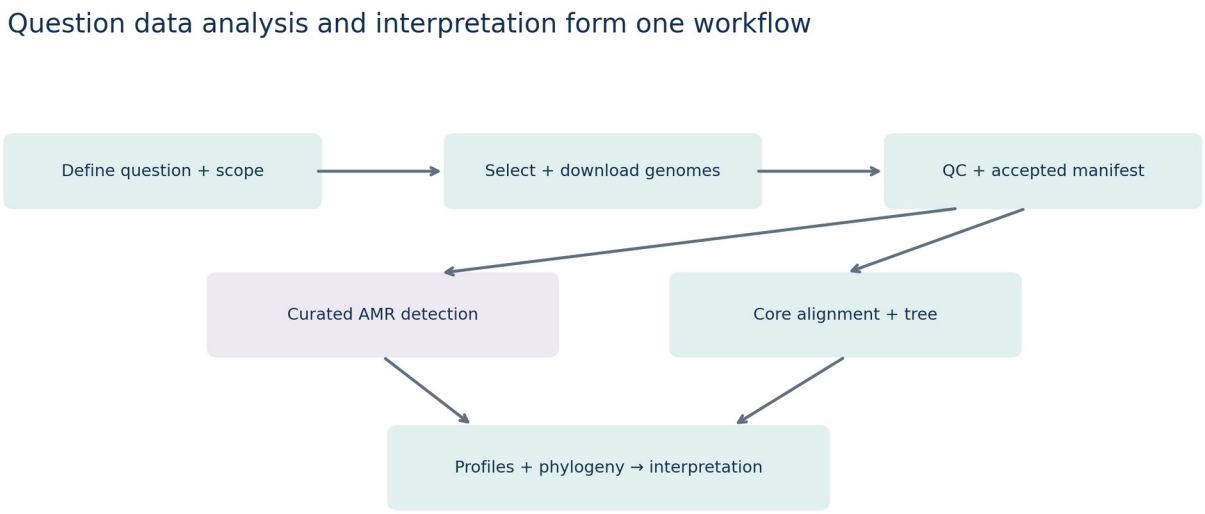


Figure 25 Figure G Complete computational workflow

Legend and interpretation. Arrows indicate data dependencies; resistance calls and phylogeny are separate evidence branches. Both branches use the same accepted isolates and converge only for interpretation.

Interpretation limit. Proposed workflow; tool output and quality gates still require human review.

Figure G is the complete analytical workflow. The biological question determines selection; quality-screened genome data feed parallel resistance and core-genome analyses; the outputs meet at interpretation. A failed QC gate should return to review rather than be converted into a negative result. This diagram helps the group explain why every step exists and which evidence supports each conclusion.

What the final results chapter should contain

Replace synthetic values only after the real analysis passes quality checks. Report the candidate, excluded and accepted counts; dataset characteristics; the evidence rules for calls; per-determinant assessable denominators; the phylogenetic method and uncertainty; and a cautious synthesis of the combined view. If no determinant is detected, report that honestly and examine detection limits. No hypothesis requires a positive finding to be educational or valid.

Chapter 16 Biological interpretation

Five scenarios to practise

Closely related strains sharing resistance genes may have inherited them from a recent common ancestor. They could also have experienced the same mobile-element acquisition or come from an oversampled outbreak. Compare allele sequences, genomic context, sampling independence and branch support before selecting an explanation. Shared content plus close ancestry does not identify a transmission chain.

Distantly related strains carrying similar genes are compatible with horizontal dissemination. However, an ancestral determinant with repeated loss, a broadly distributed gene family, convergence, contamination or uncertain topology may also explain the observation. Stronger evidence includes matching alleles and flanking regions on resolved mobile elements. A family-level heatmap alone is insufficient.

An isolate with multiple determinants may have potential to resist several antibiotic classes, or it may carry partly redundant genes affecting the same class. Some genes may be disrupted or weakly expressed. Summarise distinct

mechanisms and validated drug associations, then state that phenotype testing would be needed to confirm resistance. Do not turn raw gene count into a clinical risk score.

Resistance genes concentrated within a lineage may reflect expansion of a successful clone, shared ecology or linkage to a mobile element maintained in that lineage. Sampling from one hospital or period can create the same pattern. Assess whether the concentration persists after obvious duplicates and closely clustered outbreak samples are recognised. A small convenience sample supports description rather than general causal claims.

Different profiles among closely related strains may reflect recent acquisition or loss, regulatory variation, assembly gaps or inconsistent detection. Start by checking sequence completeness, raw calls and annotation compatibility. Biological interpretation should follow technical review, not replace it. Nearby tree tips need not have identical accessory genomes.

From observation to defensible claim

Use a three-part sentence: state the observation, explain a plausible mechanism, then identify the remaining uncertainty. For example: In the selected genomes, a supported core-genome clade carried the same ARG family; this is consistent with shared inheritance or acquisition within the lineage; allele context and broader sampling are required to distinguish these explanations. This structure separates what was measured from what was inferred.

A negative association is also informative within limits. If profiles vary greatly within a clade, ancestry alone may not explain content. That does not prove selection has no effect or that all genes transfer freely. Our conclusion should answer the research question at the resolution actually available: selected genomes, curated determinants, a specified core alignment and explicit uncertainty.

Chapter 17 Research limitations

Selection metadata and measurement

Public genomes are a convenience sample shaped by sequencing capacity, research priorities and outbreak investigation. The study cannot estimate global or Kenyan E. coli prevalence from an unrepresentative collection. Missing source, host, date and location metadata limit comparison; complete-case restriction may introduce further bias. Document missingness rather than imputing plausible-looking facts.

Assemblies can omit plasmids, fragment genes or contain contamination. A non-detection therefore means no qualifying hit under the chosen method and data quality, not proof that the source bacterium lacked every resistance mechanism. Complete assemblies reduce some uncertainty but are not error-free and may be selectively available from certain settings.

Detection depends on the tool and reference snapshot. Known genes are better represented than novel mechanisms. Different similarity rules, partial-gene handling and curated mutation catalogues can change the profile. Cross-tool agreement provides supporting evidence, not independent laboratory confirmation. Overexpression, regulatory changes, gene dosage and interactions may be missed by a simple gene list.

Evolution phenotype and inference

Genotype does not fully determine phenotype. Laboratory conditions, expression and incomplete knowledge affect prediction. This study does not measure MICs or validate therapy recommendations. If public AST data are added, ensure they belong to the same isolate and retain method, drug, units, breakpoint system and version; heterogeneous historical interpretations are not directly interchangeable.

Horizontal transfer remains uncertain without suitable genomic context. Short contigs may fail to resolve whether a determinant is chromosomal or plasmid-borne. A shared ARG is not proof of direct exchange. Trees have uncertainty

from alignment error, recombination, model assumptions, root selection and limited informative sites. A species-level tree is not necessarily the history of each ARG.

Cross-sectional data cannot establish a historical increase, direction of transfer, exact acquisition date or selection pressure experienced by the organism. Dated samples alone do not guarantee adequate temporal signal. Small collections also limit formal inference, and phylogenetic dependence makes naive statistical tests misleading.

How to report limitations constructively

For each limitation, connect it to the affected claim and a mitigation. For example: fragmented assemblies can produce false non-detections, so inspect partial hits and preserve uncertain calls; the remaining limitation is incomplete recovery of gene context. This is more useful than a generic statement that more research is needed. The project is successful if its outputs are reproducible, its biological interpretations are coherent and its claims remain proportional to the evidence.

Chapter 18 Key terminology and glossary

The following 60 terms are intended for quick reference. Read definitions in the context of the full chapters; a short label does not replace evidence or a method description.

Accessory genome. Genes variably present among the genomes in a defined collection; a plasmid ARG may be accessory.

Acquired resistance. Resistance arising through newly acquired DNA or heritable change rather than a usual species property.

Allele. A sequence variant of a gene; family-level detection does not always resolve the allele.

Alignment. An arrangement of homologous sequences into comparable columns.

AMR. Antimicrobial resistance; antibiotic resistance is the antibacterial subset relevant here.

Annotation. Computational identification and labelling of genomic features and likely functions.

ARG. Antibiotic resistance gene whose product can contribute to resistance in a suitable context.

Assembly. A reconstruction of sequence from reads; it may contain many contigs.

Ascertainment bias. Distortion caused by how sites or samples were selected; SNP-only alignments omit constant sites.

Base pair. A complementary pair of DNA bases; A–T or C–G.

Binary fission. Bacterial division producing two descendants after genetic replication and separation.

BioSample. A database record describing sampled biological material, distinct from an assembly or read run.

BLAST. A tool for local sequence similarity searches; a hit is not automatic functional confirmation.

Bootstrap support. A measure of branch stability under a resampling procedure, not a transmission probability.

Chromosome. The principal genomic replicon; E. coli usually has one main circular chromosome.

Clade. An ancestor and all descendants represented by that part of a rooted tree.

Clonal expansion. Increase in descendants of a lineage, potentially spreading inherited resistance.

Coding sequence. DNA region translated into protein in a specified reading frame.

Codon. Three nucleotides interpreted together during translation; TTT encodes phenylalanine in the standard code.

Conjugation. Contact-dependent DNA transfer mediated by transfer machinery.

Contig. A contiguous assembled sequence; it is not necessarily an entire chromosome.

Core genome. Features shared across a collection under an explicit presence definition.

Coverage. Can mean sequence depth or fraction of a reference matched; always state which.

Efflux. Transport of compounds out of the cell; some pumps reduce intracellular antibiotic exposure.

ESBL. Extended-spectrum beta-lactamase; the label requires appropriate enzyme/variant evidence.

FASTA. A labelled sequence format with greater-than-sign headers; no per-base quality scores.

FASTQ. A sequence-read format including encoded per-base quality scores.

Fitness. Relative reproductive success in a specified environment.

Gene. A DNA region contributing a functional RNA or protein product.

Gene cassette. A genetic unit that can be captured by an integron; not a synonym for an entire plasmid.

Genome. The full genetic complement, including relevant extrachromosomal DNA.

Genotype. The genetic constitution of an organism or a defined locus.

GFF3. A structured text format for feature coordinates and attributes.

Homolog. A sequence related to another through shared evolutionary ancestry.

Horizontal gene transfer. Transfer of genetic material outside ordinary parent-to-descendant inheritance.

Indel. An insertion or deletion relative to a comparison sequence.

Integron. A system that captures and expresses gene cassettes; it is not generally self-transmissible.

Intrinsic resistance. A resistance property associated with usual species features.

Isolate. Biological material from a specific sampling event; distinct records can represent the same isolate.

Jaccard similarity. Shared positive features divided by the union of positive features.

MIC. Minimum inhibitory concentration measured under standardised conditions.

Mobile genetic element. A genetic element capable of movement within or between genomes by a particular mechanism.

Mutation. A heritable sequence change; its effect depends on context.

N50. A length statistic: contigs of this length or longer cover at least half the assembly length.

Nonsynonymous substitution. A coding change that alters an amino acid in the specified reading frame.

Ortholog. A homolog descended through lineage divergence rather than gene duplication.

Pangenome. The union of gene families across the analysed collection.

Paralog. A homolog arising through gene duplication; duplicate copies can complicate core inference.

Phenotype. An observable property under defined conditions, such as a measured MIC.

Phylogeny. An inferred evolutionary relationship among organisms or sequences.

Plasmid. An independently replicating DNA molecule; some carry ARGs or transfer functions.

Porin. An outer-membrane channel that can affect entry of certain molecules.

Promoter. A regulatory DNA region involved in initiating transcription.

Recombination. Genetic exchange that rearranges or replaces sequence, including homologous exchange.

Resistome. The set of resistance determinants considered in an organism, community or environment.

Selection pressure. An environmental condition that changes relative survival or reproductive success.

SNP. A single-nucleotide difference at a comparable genomic position.

Strain. A distinguishable lineage or variant within a bacterial species.

Transformation. Uptake and stable establishment of extracellular DNA by a competent cell.

Transduction. Bacteriophage-mediated movement of bacterial DNA.

Chapter 19 Group study and discussion questions

Questions from foundations to critique

1. Why is antibiotic resistance a bacterial property rather than a patient property?

2. Name three structures in an E coli cell relevant to this project.

3. How does an isolate differ from a strain?

4. Explain the difference between DNA a gene and a protein.

5. What is the complementary strand to 5′ ATGC 3′?

6. Why is plasmid presence not equivalent to resistance?

7. What distinguishes beta lactamase activity from a porin change?

8. Why is a gyrA hit insufficient to call fluoroquinolone resistance?

9. Does an antibiotic cause the exact mutation a bacterium needs?

10. Can resistance frequency rise while bacterial numbers fall?

11. Contrast vertical inheritance with conjugation.

12. Why should an integron not be described as a conjugative plasmid?

13. What data are stored in FASTA and FASTQ?

14. What is wrong with choosing all genomes from the first results page?

15. Why does a high N50 not prove that an assembly is correct?

16. How do core accessory and pangenome differ?

17. Compute frequency when six of eight assessable isolates carry a gene and two are unassessable.

18. Why should phylogeny not be built from the same ARG matrix used to test lineage clustering?

19. What does a bootstrap support value establish?

20. Can a tree alone identify the donor of a shared resistance plasmid?

21. Why could two close tree tips have different resistance profiles?

22. What should happen when AMRFinderPlus and another resource disagree?

23. Why can a cross sectional collection not establish a historical rise in resistance?

24. A human and an animal isolate share blaCTX M What conclusion is justified?

25. What is the minimum evidence package for a reproducible project?

Model answers

1. Bacteria carry mechanisms that reduce the effect of a drug under defined conditions. The patient may carry or be infected by resistant bacteria; the patient does not become resistant to the antibiotic in the same sense.

2. The chromosome carries inherited genes, plasmids may carry accessory resistance determinants, and the envelope affects drug entry. Ribosomes are both protein-production machinery and antibiotic targets.

3. An isolate refers to a sampling event/material. A strain is a distinguishable lineage. Several isolates or database assemblies may represent one strain or the same original sample.

4. DNA is the sequence-bearing molecule. A gene is a functional genomic region. A protein is an amino-acid product translated from a protein-coding gene through an RNA intermediate; some genes produce functional RNA instead.

5. It is 3′ TACG 5′ when aligned antiparallel. Written in its own 5′ to 3′ direction, the reverse complement is 5′ GCAT 3′.

6. Many plasmids carry no ARG. An ARG must be present, intact and functionally relevant in context, and even that supports a prediction rather than direct phenotype measurement.

7. A beta-lactamase chemically inactivates susceptible beta-lactams. A porin change alters entry and exposure. Both can contribute together but are different mechanisms.

8. gyrA is a normal topoisomerase gene. The relevant evidence is a supported, validated substitution or combination, not presence of the gene itself.

9. No purposeful targeting is implied. Selection changes survival or reproduction of variants. Stress can affect mutation dynamics, but it does not direct a precise adaptive sequence change.

10. Yes. If susceptible cells decline faster, resistant cells can form a larger fraction of a smaller population. Relative frequency and absolute abundance are different measurements.

11. Vertical inheritance passes genetic material to descendants through reproduction. Conjugation transfers DNA between contacting cells through transfer machinery and can be followed by vertical inheritance.

12. An integron captures and expresses gene cassettes. A plasmid is a replicating molecule and may carry transfer functions. Integrons may be embedded within mobile elements but are not generally self-transmissible.

13. FASTA stores labelled sequences. FASTQ stores reads plus encoded per-base quality scores. Neither file extension alone proves that a sequence is complete or correctly assembled.

14. Database ranking and availability can create a convenience sample with outbreak, geography or study biases. Eligibility and sampling rules should be defined before examining resistance profiles.

15. N50 describes contiguity. A misassembled or contaminated sequence can still be long, and plasmids or genomic regions can remain missing. Other quality and metadata checks are necessary.

16. Core features are shared under the chosen rule; accessory features vary; the pangenome is their union. All depend on the sampled genomes and clustering definitions.

17. The assessable frequency is 6/8 = 75%, with two NA out of ten selected isolates. Reporting 6/10 without explaining missingness would imply the two unassessable isolates were negatives.

18. That would make the relationship partly circular. Infer ancestry using independent core sequence information, then map ARG content onto those relationships.

19. It describes branch stability under the stated resampling method. It does not give the probability of direct transmission, provide an acquisition date or eliminate model and sampling bias.

20. No. Gene context, plasmid resolution, epidemiological information and sampling of alternatives are needed, and even these may leave donor direction unresolved.

21. Accessory-gene gain or loss, different plasmids, regulatory changes, incomplete assemblies or detection inconsistencies can produce the difference. Technical review should precede a biological claim.

22. Check input type, database version, match coverage, identity, allele naming, model category and partial-gene rules. Preserve both raw calls and report the reason for the final interpretation or remaining uncertainty.

23. It lacks a suitable temporal design and potentially representative dated sampling. Differences among present-day lineages are not equivalent to an increase through calendar time.

24. The selected isolates share a determinant family. This may motivate a One Health dissemination hypothesis, but it does not establish direction, direct exchange, a shared allele or a particular reservoir.

25. Accepted/excluded manifests, accession versions, metadata, input checksums, software/database versions, commands, logs, raw calls, transformations, alignment/tree files, plot data and cautious conclusions linked to objectives.

Chapter 20 Literature review and references

How the literature supports this project

The literature review should connect three levels of explanation. Molecular studies and reviews establish how a determinant can alter antibiotic action. Population genetics explains why the same determinant can appear within a successful clone or across different lineages. Tool validation and comparative-genomic methods explain how sequence data can support those claims and where error enters.

Davies and Davies (2010) provide historical and evolutionary framing. Tenaillon et al. (2010) explain E. coli population structure and commensal ecology. These are foundational rather than a description of the present genome database. Munita and Arias (2016) organise resistance mechanisms, while Didelot and Maiden (2010) explain the importance of recombination. Read these sources as complementary explanations, not as evidence that the selected project genomes have already shown the same patterns.

Feldgarden et al. (2021), Alcock et al. (2023) and Bortolaia et al. (2020) describe resistance analysis resources with distinct scopes and validation approaches. Their tools should not be treated as identical black boxes. Gurevich et al. (2013), Schwengers et al. (2021), Tonkin-Hill et al. (2020) and Minh et al. (2020) support assembly assessment, annotation, pangenome analysis and tree inference respectively. Original method papers explain rationale; current official documentation is needed for executable commands and release-specific behaviour.

Building a focused review

Create an evidence table recording the question, organism, number and source of isolates, sequencing approach, resistance method, phylogenetic method, key interpretation, limitations and relevance to our objectives. Read the methods before quoting a headline. A study using matched susceptibility measurements can assess genotype–phenotype agreement, whereas a genome-only study cannot supply that validation by itself.

Search Google Scholar using combinations such as Escherichia coli comparative genomics acquired resistance core genome phylogeny; E coli plasmid blaCTX-M horizontal gene transfer genomic context; AMRFinderPlus genotype phenotype validation Escherichia; bacterial phylogeny recombination core alignment ascertainment bias; and One Health E coli resistance genomic surveillance sampling bias. Add a date filter for recent literature and trace both cited and citing papers. Search strings are recommendations, not claims that the searches have already identified a final eligible literature set.

References in APA style

References below were checked against discoverable publisher, PubMed or official project records while preparing this handbook. Software and live database documentation should be revisited when the actual analysis begins. DOI links identify publications; a tool's current release is recorded separately in the analysis manifest. Long author lists follow APA 7 truncation after the first 19 authors and before the final author where required.

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Animation companion and narration

The accompanying GIFs are separate playable files, each with a static counterpart in this handbook. Word playback varies by version, so the handbook does not depend on embedded animation. These are conceptual teaching animations, not simulations fitted to experimental data.

01 antibiotic selection

Static counterpart Figure on natural selection. Scene 1 shows mostly green susceptible cells and a few orange variants. Scene 2 removes susceptible cells during exposure. Scene 3 shows orange descendants expanding. Explain that survival and reproduction alter population composition; no cell deliberately chooses a beneficial mutation. Shapes are conceptual rather than quantitative survival estimates.

02 plasmid transfer

Static counterpart Conjugation figure. A donor retains an orange plasmid symbol while another DNA symbol moves along a connection to the recipient. Narrate contact, DNA processing/transfer and establishment. The moving symbol stands for transferred DNA; it is not a literal intact circle travelling through a hollow pilus. Both cells may retain the element.

03 nucleotide substitution

Static counterpart Sequence alignment figure. The displayed coding fragment begins with TTT at the selected codon. One highlighted base changes to give TCT, and the translated label changes from phenylalanine to serine. The example demonstrates a nonsynonymous change without claiming that the toy fragment is a real ARG or that this change confers resistance.

04 phylogenetic branching

Static counterpart Hypothetical phylogeny figure. Branches are progressively revealed from a teaching root to four labelled tips. Pause at internal nodes to identify shared ancestors. The order of reveal is a teaching device, not an estimate of elapsed time or observed evolution. The full handbook tree uses ten isolates; this animation simplifies the same branching concept.

05 analysis workflow

Static counterpart Complete workflow figure. Five stages are highlighted in turn: question/selection, acquisition, quality assessment, AMR and core-tree analysis, and interpretation. Explain the data dependencies and return to a prior stage when QC fails. Highlighting does not mean any project analysis has been executed.